Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: Tam, Jayden WA3013088

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a =$ 27.03 $\quad b =$ 1.67 $\quad r =$ 0.67

Subtract 1 from the Growth factor

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

(a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

$a = 9$

Convert the inflation rate into a decimal $r = \frac{4.1}{100} = 0.041$

Exponential growth formula - $P(t) = a(1+r)^t$

$P(t) = 9(1+0.041)^t$

$P(t) = 9(1.041)^t$

(b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent. $t = 25$

$P(25) = 9(1.041)^{25}$

$\approx \$24.57$

The movie ticket will cost approximately \$24.57 in 25 years assuming inflation stays at 4.1% annually.

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

(a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t. Using the formula $P(t) = P_0 + mt$ where $P_0 = 6500$ and $m = -60$

$P(t) = 6500 - 60t$

(b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

Use the formula $P(t) = a(1-r)^t$ where $a = 6500$

$r = 0.05$ (5% to 0.05)

$1 - 0.05 = 0.95$

$P(t) = 6500(0.95)^t$

4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

Since $0 < b < 1$, the exponential term approaches 0 as $t \to \infty$

$b^t \to 0$ Therefore $y = ab^t \to a(0)$

$y = 0$

5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$

Look at the base of 0.75

Since $0 < 0.75 < 1$, the exponential term approaches 0 as $x \to \infty$

$(0.75)^x \to 0$

Multiply by 192 $192(0) = 0$

(b) (4 pts) As $t \to -\infty$, $2(3)^t \to 0$

base is $3 > 1$

$(3)^t \to 0$

$2(0) = 0$

When t approaches negative infinity, the exponential term approaches 0

(c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to 15$

Since $0 < 0.9 < 1$, this term approaches to 0

$(0.9)^t \to 0$

$5(3+0) = 15$

(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to 0$

Look at the exponent $0.5t$

As $t \to -\infty$, $0.5t \to -\infty$

Therefore, $e^{0.5t} \to 0$

Multiply by 8

$8(0) = 0$

6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.

At the intersection $f(t_0) = g$

So, $a(1+r)^{t_0} = b(1+s)^{t_0}$

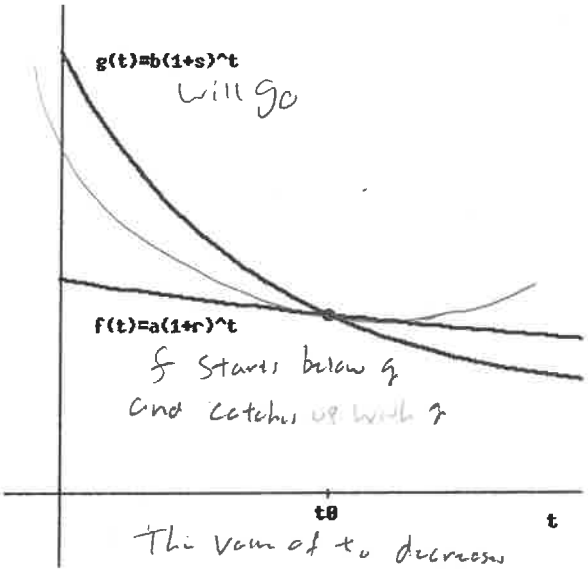
$\frac{a(1+r)^{t_0}}{a(1+s)^{t_0}} = \frac{b}{a}$

$\left(\frac{1+r}{1+s}\right)^{t_0} = \frac{b}{a}$

$\ln\left[\left(\frac{1+r}{1+s}\right)^{t_0}\right] = \ln\left(\frac{b}{a}\right)$

$t_0 \ln\left(\frac{1+r}{1+s}\right) = \ln\left(\frac{b}{a}\right)$

$t_0 = \frac{\ln\left(\frac{b}{a}\right)}{\ln\left(\frac{1+r}{1+s}\right)}$



The value of t_0 decreases

Increasing s makes $f(t)$ stay unchanged while $g(t)$ is gro
Therefore $g(t)$ reaches $f(t)$ sooner, so the intersection
moves to the right and the value of t_0 decreases

7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$m(0) = 160e^{-0.025(0)}$

$m(0) = 160e^{0}$ $\qquad e^0 = 1$

$m(0) = 160(1)$

$= 160$ grams

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

$m(10) = 160e^{-0.025(10)}$

$m(10) = 160e^{-0.25}$

Multiply the exponents $\qquad -0.025(10) = -0.25$

$m(10) \approx 124.6081$ grams

$160e^{-0.25} \approx 124.6081$

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

   (a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order. Transformation 1 - Moves the Graph 4 units to the left

   $e^{x} \xrightarrow{-4} e^{x}$

   Transformation 2 - Shifts 2 units up

   (b) (4 pts) State the domain and range of $f(x)$.

   Domain - $(-\infty, \infty)$

   Range = $(2, \infty)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

   $\ln A = B \longleftrightarrow e^B = A$

   (a) (4 pts) $\ln 10 = y$

   The base of ln is e

   $e^y = 10$

   (b) (4 pts) $\ln x = 5$

   $e^5 = x$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

   (a) (4 pts) $\ln(e^{1.7})$

   Since ln and e cancel out, the exponent remains

   $\ln(e^{1.7}) = 1.7$

   (b) (4 pts) $e^{\ln 4}$

   $e^{\ln 4} = 4$ Since e and ln cancel out, what is inside the logarithm remains

   (c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

   $\ln\left(\frac{1}{e^3}\right)$

   $\frac{1}{e^3} = e^{-3}$ Rewrite the fraction using a negative exponent

   $\ln(e^{-3})$ Substitute into expression

   $= -3$ Cancel ln and e

   exponent left

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

$\ln(e^{-3x+2}) = \ln 10$

$-3x+2 = \ln 10$

$-3x+2-2 = \ln 10 - 2$

$\frac{-3x}{-3} = \frac{\ln 10 - 2}{-3}$

$x = \frac{\ln 10 - 2}{-3}$

(b) (4 pts) $8 - \ln(2-x) = 1$.

$8 - \ln(2-x) - 8 = 1 - 8$

$-\ln(2-x) = -7$

$(-1)[-\ln(2-x)] = -7(-1)$

$\ln(2-x) = 7$

$e^7 = 2-x$

$e^7 - 2 = 2 - x - 2$

$e^7 - 2 = -x$

$(-1)(e^7 - 2) = (-1)(-x)$

$-e^7 + 2 = x$ So, $x = 2 - e^6$

(c) (4 pts) $4^x = 3$

$\ln(4^x) = \ln 3$

$\frac{x \ln 4}{\ln 4} = \frac{\ln 3}{\ln 4}$

$x = \frac{\ln 3}{\ln 4}$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$\frac{2 \cdot 4^{2x-4}}{2} = \frac{60}{2}$

$4^{2x-4} = \frac{60}{2}$

$\ln(4^{2x-4}) = \ln(\frac{60}{2})$

$\frac{(2x-4)\ln 4}{\ln 4} = \frac{\ln(\frac{60}{2})}{\ln 4}$

$2x - 4 = \frac{\ln(\frac{60}{2})}{\ln 4}$

$2x - 4 + 4 = \frac{\ln(\frac{60}{2})}{\ln 4} + 4$

$\frac{4x}{4} = \frac{\ln(\frac{60}{2}) + 4}{4}$

$x = 1 + \frac{\ln(\frac{60}{2})}{4\ln 4}$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured. We know that at $t = 2$ minutes, the coffee cooled to 190°

a = Initial Temperature

c = Room temp (75)

$115 + 75 = 190$

$F(t) = 115e^{-k2} + 75$

$190 = 115e^{-k2} + 75$

$190 - 75 = 115e^{-k2}$

$\frac{115}{115} = \frac{115e^{-k2}}{115}$

$\frac{115}{115} = e^{-k2}$

$\ln\left(\frac{115}{115}\right) = \ln e^{-k2}$

$\ln\left(\frac{115}{115}\right) = -k2$

$\frac{\ln\left(\frac{115}{115}\right)}{-2} = \frac{-k2}{-2}$

$k = \frac{\ln\left(\frac{115}{115}\right)}{-2} \simeq 0$

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: Emmanuel Cruzman WA3285729

Please read the following instructions carefully.

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- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a = 27.03$ $b = 1.67$ $r = 0.67$

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

$P(t) = ab^t$

$P(t) = 12.75(1+0.041)^t$

$P(t) = 12.75(1.041)^t$

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

$P(25) = 12.75(1.041)^{25}$

$P(25) = \$26.025$

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

$P(t) = y_0 + m(t - t_0)$

$m = -60$

t_0 y_0

$(0, 6500)$

$P(t) = 6500 - 60t$

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

$P(t) = ab^t$

$a = 6500$

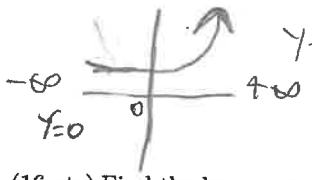
$r = -0.05$

$b = 1 + r = 1 - 0.05 = 0.95$

$P(t) = 6500(0.95)^t$

4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

$0<b<1$ $y=ab^t$ as $t \to -\infty$



no we do not have a horizontal asymptote, It is vertical

$f(x)=ab^x$

5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$ 0

as $x \to \infty$, $f(x) = 192(0.75)^x$

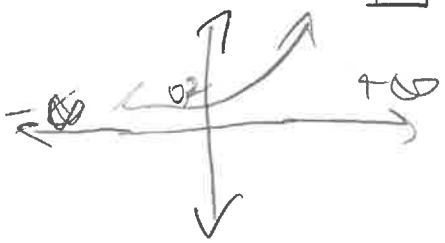
$f(x) =$



$0 = b = 0.75$ B

approaches to 0

(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$ 0

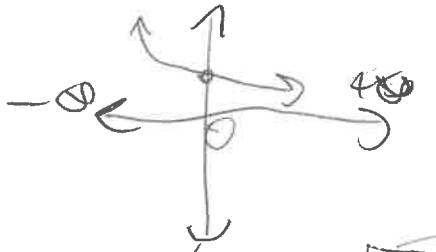


As $t \to -\infty$ $f(x) = 2(3)^t \to 0$

$B = 3$ 11

(c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to$

As $t \to \infty$, $F(4) = 5(3+(0.9)^4) \to 20.45$



$= f(4) = (1.0.9)^t$

$= 5(3 + 1(.09)$

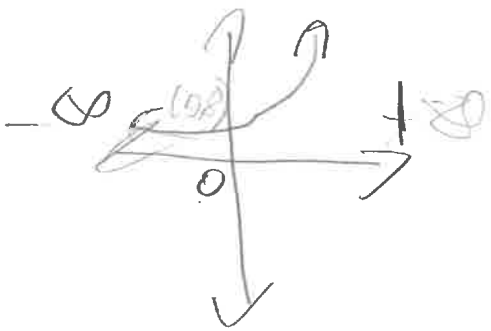
$= 5(4.09)$

$= 20.45$

(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$ 0

as $t \to -\infty$ $f(x) 8e^{0.5t} \to 0$

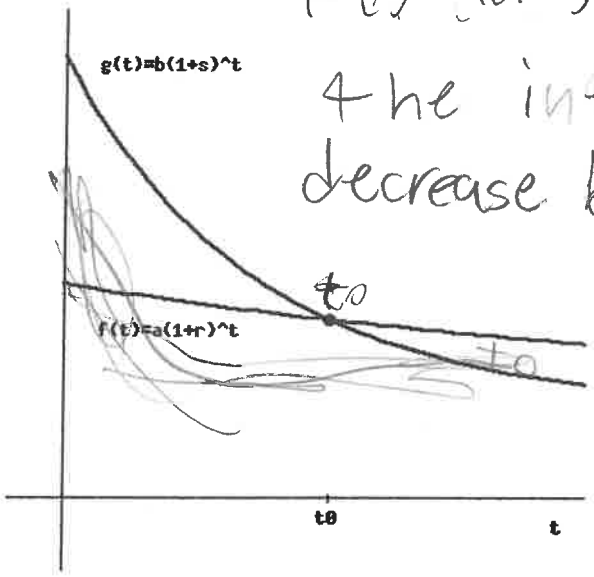
$B = > 1$



6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.

$f(t)=a(1+r)^t$ and $g(t)=b(1+s)^t$

the intersection would decrease but increase



7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$m(0) = 160e^{-0.025(0)}$

$= 160e^{0}$

$= 160(1)$ $(m(0) = 160$ grams$)$

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

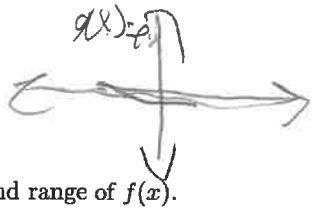
$t = 10$

$m(10) = 160e^{-0.025(10)}$

$m(10) = 1$

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

   (a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.



   (b) (4 pts) State the domain and range of $f(x)$.

Domain : $(-\infty, \infty)$

Range : $(2, \infty)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

   (a) (4 pts) $\ln 10 = y$

$e^y = 10$

   (b) (4 pts) $\ln x = 5$

$e^5 = x$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

   (a) (4 pts) $\ln(e^{1.7})$

$= 1.7$

   (b) (4 pts) $e^{\ln 4}$

$= 4$

   (c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

$= \ln(e^{-3})$

$= -3$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

$e^{-3x+2} = 10$

$\ln(e)^{-3x+2} = \ln 10$

(b) (4 pts) $8 - \ln(2 - x) = 1$.

(c) (4 pts) $4^x = 3$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: Harrington, Michael WA3262256

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- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a = 27.03$ $\quad b = 1.67$ $\quad r =$ _____

27.03 $\quad (1.67)^t$ $\quad q = ab^t = a(1+r)^t$ $\quad q = ab^t = a(1+t)^t$ $\quad 27.03(1+t)^{1.67}$

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

$12.75(t) = 12.75^t(0.41^t)$ t = ticket

$\left[P(t) = \frac{A}{4.1 + me^{-kt}}\right]$

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

$[130.7]$

$P(t) = 25(t)$

$P(t) = -t(x)$

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

$P(t) = ab^t$ $\quad P(t) = ab^t$ $\quad t = 0$ $\quad P = 60$ $\quad 0 = ab^0$ $\quad 60 = ab^{6500}$ $\quad 6500 = ab^{60}$ $\quad 60 = ab^0$

$A =$

$V(t) = ab^t$

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

$P(t) = ab^{0.5}$

P

$6500(t) = 6500^{0.5}$

4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

$0<b<1$ $\quad y=ab^{t}$

$y=ab^{+}$ $\quad +\Rightarrow \infty$

?

5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$

$x \Rightarrow \infty$ x

$192(0.75)^{+} \Rightarrow$

-141^{+}

(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$

$+ \Rightarrow -\infty$

$2(3)^{+} \Rightarrow (6^{+})$

-18^{+}

(c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to$

$+ \Rightarrow \infty$

$5(3+(0.9)^{+}) \Rightarrow$

63^{+}

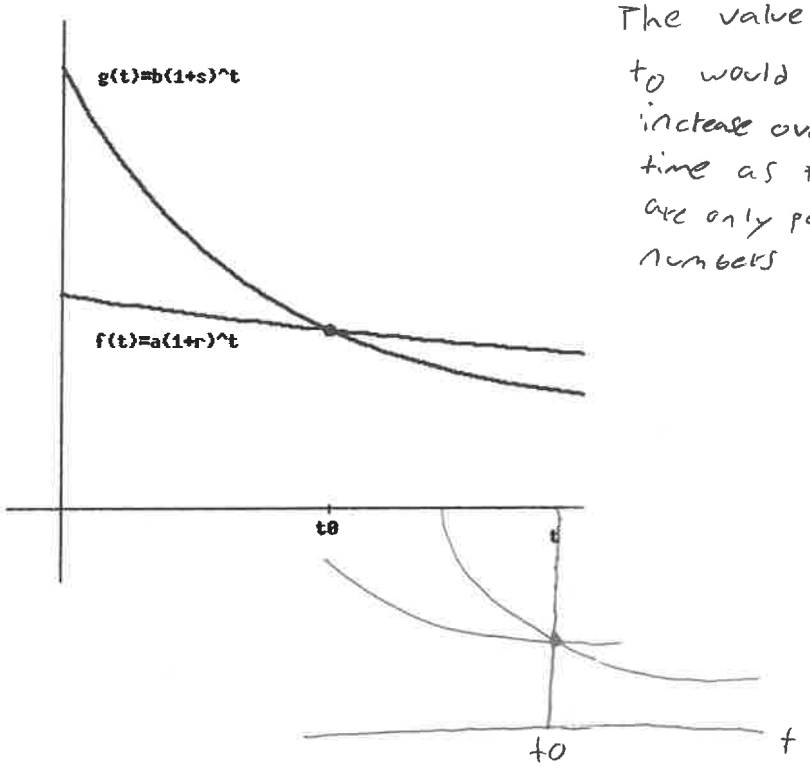
(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$

$+ \Rightarrow -\infty$

$8e^{0.5+} \Rightarrow$

6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.

The value of t_0 would increase over time as there are only positive numbers



7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

[-4] $m(t) = 160e^{-0.025(0)}$

$160e$

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

[-40]

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

   (a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

   (b) (4 pts) State the domain and range of $f(x)$.

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

   (a) (4 pts) $\ln 10 = y$

   $e^A = B$

   $\ln 10 = y$

   $\ln 10 = y \quad \ln = \frac{\ln 10}{y}$

   (b) (4 pts) $\ln x = 5$

   $\ln x = 5 \qquad \ln = \frac{\ln x}{\ln 5}$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

    (a) (4 pts) $\ln(e^{1.7})$

    $\ln(e^{1.7})$

    (b) (4 pts) $e^{\ln 4}$

    $e^{\ln 4}$

    (c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

    $\ln\left(\frac{1}{e^3}\right)$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

$e^{-3x+2} = 10$

$e(2.5)^{3(-7)+2} = 10e$

(b) (4 pts) $8 - \ln(2 - x) = 1$.

$8 - \ln(2-x) = 1$

$\ln(ab) = \ln(8) - \ln(2)$

$8^x - 2 = 1$

$(1)\ 8^x(-2) = 1$

$\log_1(x) = \frac{\ln -2(x)}{\ln 8(1)}$

(c) (4 pts) $4^x = 3$

$4^x = 3$

$4^{(\ln 3)} = 3$

$4^{(0.25)}$

$4^{(0.75)} = 3$

$[x = 0.75]$

$\ln(ab) = \ln(4) + \ln(3)$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$2 \cdot 4^{2x-4} = 60$

$2 \cdot 4^{(2x-4)} = 60$

$64^{(x-4)}$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

$200^{\circ}(t)$

$200^{\circ}(t) = ae^{-Kt} + C$

$F(0) = 200$

$F(0) = ae^{-Kt} + C$

$F(0) = a + C$

So $a + C = 200$

$t = 7\infty, F(t) = 75$

$t = 7\infty, f(t) = 7$

$f(t) = ae^{-Kt} + C$

$= \frac{a}{e^{Kt}} + C$

$\searrow 0$ as $t = 7\infty$

$C = 75$

$A = 125$

$\frac{\ln\left(\frac{120}{125}\right)}{-2} = \frac{-K2}{-2}$

$\left[K = \frac{\ln\left(\frac{120}{125}\right)}{-2} = \right] 0.48$

$t = \frac{\ln\left(\frac{74}{125}\right)}{\ln\left(\frac{120}{125}\right)} \times 2$

$F(2) = 190$

$F(2) = ae^{-K(2)} + C$

$190 = 125e^{-K2} + 75$

$190 - 75 = 125e^{-K2}$

$\frac{120}{125} = \frac{125^{-K2}}{125}$

$\frac{120}{125} = e^{-K2}$

$\ln\left(\frac{120}{125}\right) = \ln e^{-K2}$

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: Hawkins III, Elijah S00022046

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

A b

$a =$ 27.03 $\quad b =$ 1.67 $\quad r =$ 1.67 − 1

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

(a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

$P_t = ab^t$

$P(t) = 12.75(1+.041)$

$P(t) = 12.75(1.041)^t$

$P = 13.27$

(b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

$P_{25} = 12.75(1+.041)^{25}$

$P = 33.18$

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

(a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

$P(t) = y_0 + m(t - t_0)$

$P_t = 6500 - 60t$

$P_t = 6500 - 60t$

$t_0 \quad y_0$

$0, 6500$

(b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

$P_t = 6500 - .05t$

$P_t = 6500(.95)^t$

$y = 6500$

$r = .05$

$b = 1 + r = 1 - .05 = .95$

4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

If $b<1$, $y=ab^t$ as $t \to -\infty$

Horizontal asymptote = $y=0$

5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$ 0 $b=.75<1$

(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$ 0 $b=3>1$

(c) (4 pts) As $t \to \infty$, $5(3+(0.9)^t) \to$ 15

$5(3+.9)$

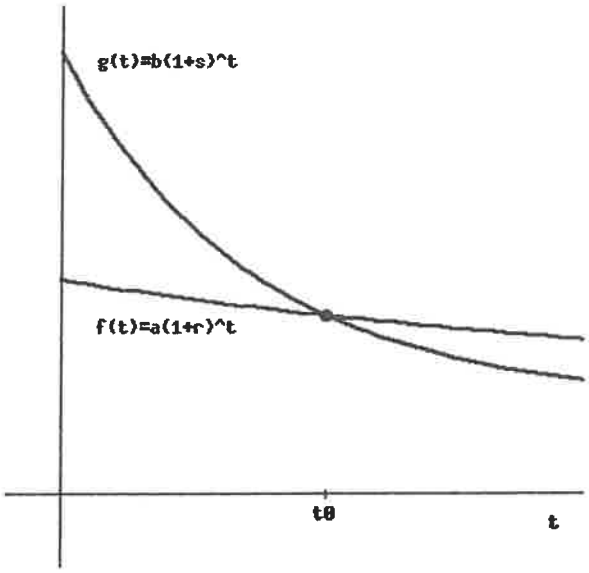
$5(3+0)$

$5(3)=15$

(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$

0

6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.



If s is increased + values a, b, r remain the same, the value of t_0 to decrease

7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$m(0) = 160$ grams

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

$m(10) = 160e^{-.025(10}$

$m(10) = 160e^{2.5}$

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

(a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

No, the order performing for these transformation doesn't matter because one transformation is horizontal and the other is vertical

(b) (4 pts) State the domain and range of $f(x)$.

Domain $= (-\infty, \infty)$

Range $= (-4, \infty)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

(a) (4 pts) $\ln 10 = y$

$\log_e 10 = y$

$\log_e 10 = y$

$e^y = 10$

(b) (4 pts) $\ln x = 5$

$\log_e x = 5$

$e^5 = x$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

(a) (4 pts) $\ln(e^{1.7})$

$= 1.7$

(b) (4 pts) $e^{\ln 4}$

4

(c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

$\log_e\left(\frac{1}{e^3}\right)$

$e^x = 1/e^3$

$e^x = e^{-3}$

$x = -3$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

$\ln(e)^{-3x+2} = \ln(10)$

$-3x+2 = \ln(10)$

$-3 = \ln(10) - 2$

$\boxed{x = \frac{\ln(10) - 2}{3}}$

(b) (4 pts) $8 - \ln(2-x) = 1$.

$-\ln(2-x) = 1-8$

$+1 \quad \ln(2-x) = -6$

$e\ln(2-x) = e^{-6}$

$2 - x = e^{-6} + 2$

$-x \quad -1 \quad -1$

$\boxed{x = e^{-6} - 1}$

(c) (4 pts) $4^x = 3^{-1}$

$\ln(4^x) = \ln(3)$

$x\ln(4) = \ln(3)$

$\frac{x\ln(4)}{\ln(4)} = \frac{\ln(3)}{\ln(4)}$

$\boxed{x = \frac{\ln(3)}{\ln(4)}}$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$\frac{2}{2} \quad \frac{60}{2}$

$4^{2x-4} = 60/2$

$\ln(4^{2x-4}) = \ln(60/2)$

$(2x-4)\frac{\ln(4)}{\ln(4)} = \frac{\ln(60/2)}{\ln(4)}$

$2x - 4 = \frac{\ln(60/2)}{\ln(4)}$

$+4 \quad +4$

$\frac{2x}{2} = \frac{\ln(60/2) + 4}{2}$

$\boxed{x = \frac{\ln(60/2) + 4}{2}}$

Hint 2 is not correct & my answer 4(1)

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

$F(0) = ae^{-k(0)} + c$

$190 = ae^0 + 75$

$190 = a + 75$ $\quad a = 190 - 75$

$\boxed{a = 115^\circ}$

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

***

Last name, first name, ID#: May, Estrella, #WA 3088321

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a =$ 25.36 $\quad b =$ _____ $\quad r =$ _____

$27.03 \; ab^t = a(1+1.67)^t$

$27.03 = a + 1.67$

$a = 27.03 - 1.67 = 25.36$

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

$4.1\%(12.75)$

$= 52.275$

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

$4.1\%(12.75)$

$= 52.275$

$52.275(25)$

$= 1306.875$

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

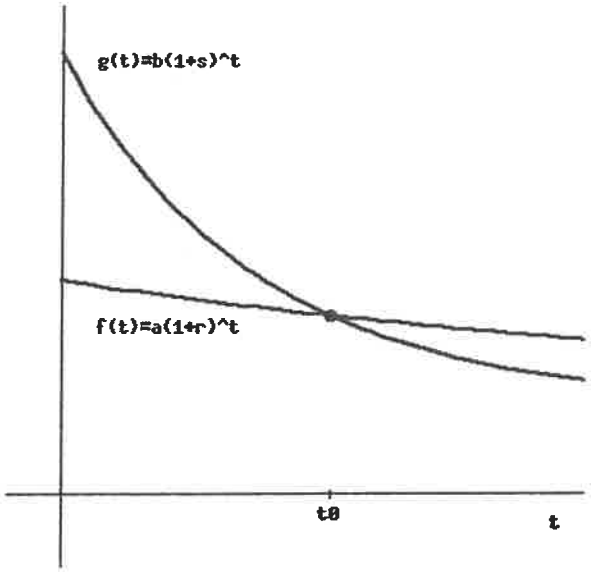
   (a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$

   (b) (4 pts) As $t \to -\infty$, $2(3)^t \to$

   (c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to$

   (d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$

6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.



7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

   (a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

   (b) (4 pts) State the domain and range of $f(x)$.

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

   (a) (4 pts) $\ln 10 = y$

   (b) (4 pts) $\ln x = 5$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

    (a) (4 pts) $\ln(e^{1.7})$

    (b) (4 pts) $e^{\ln 4}$

    (c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

(b) (4 pts) $8 - \ln(2 - x) = 1$.

(c) (4 pts) $4^x = 3$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: Gonzalez, cristhian, WA3009254

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a = 27.03$ $b = 1.67$ $r = 0.67$

$Q = ab^t = a(1+r)^t$

$Q = 27.03(1.67)^t$ $r = 1.67 - 1 = 0.67$

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

$P(t) = a(1+r)^t$

$A = 12.75$

$r = 0.041$

$P(t) = 12.75(1.041) = 13.27275$

the ticket started at 12.75 A yearly increase of 4.1% means multiply of cost by 1.041 each

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

$t = 25$

$P(25) = 12.75(1.041)^{25} = 33.181875$

13.27275 in one year

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

6500 $P(0)$ $P(t) = 6500 - 60^t$

$P(0) = 6500 - 60$

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

$100\% - 5\% = 95\%$ $95 = 0.95$

$P(t) = 6500(0.95)^t$

4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

$y = ab^t$ $\quad b > 1$

$b^t = \frac{1}{b^{-t}}$ $\quad t \to \infty, \; t \to -\infty$

$y = 0$

5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$ (b)

$0 < 0.75 < 1$ $\quad (0.75)^x \to 0$

$192(0.75) \to 192(0) = \boxed{0}$

(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$

$(3)^t = \frac{1}{(3)^{-t}} = \boxed{0}$

(c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to$

$5(3+0) = 5(3) = \boxed{15}$

$0.9^t = \frac{1}{0.9^{-t}} = 0$

(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$

$t \to -\infty$, the exponent $0.5t \to -\infty$ and 0.5^t

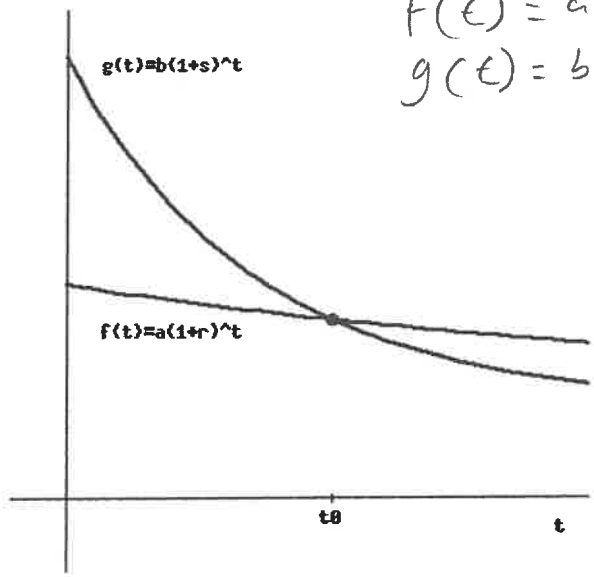
$e > 0$

$8e^{0.014^t} \to 8(e) = \boxed{0}$

6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.

$f(t) = a(1+r)^t$

$g(t) = b(1+s)^t$



7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$m(0) = 160e^{-0.025(0)} = 160^0 = 160(1) = 160$

160 grams

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

$m(10) = 160e^{-0.025(10)}$

$m(10) = 160e^{-0.25} = 2.8117$

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

   (a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

   $f(x) = e^{x-4} + 2$

   $g(x) = e^x$

   (b) (4 pts) State the domain and range of $f(x)$.

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

   $\ln A = B \iff e^A = B$

   (a) (4 pts) $\ln 10 = y$

   $e^y = 10$

   (b) (4 pts) $\ln x = 5$

   $\ln x = 5$

   $e^5 = x$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

    (a) (4 pts) $\ln(e^{1.7})$

    1.7

    (b) (4 pts) $e^{\ln 4}$

    4

    (c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

    $\ln\left(\frac{1}{e^3}\right) = e^{-3}$

    $\ln(e^{-3}) = -3$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

$$\ln\left(e^{-3x+2}\right) = \ln 10$$

$$-3x + 2 = \ln 10$$

$$x = \frac{\ln 10 - 2}{-3}$$

(b) (4 pts) $8 - \ln(2 - x) = 1$.

$$\ln(2-x) = 8$$

$$2 - x = e^8$$

$$-x = e^8 - 2$$

$$x = 2 - e^8$$

(c) (4 pts) $4^x = 3$

$$\ln\left(4^x\right) = \ln 3$$

$$x \ln 4 = \ln 3$$

$$x = \frac{\ln 3}{\ln 4}$$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$$\frac{1}{128}$$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

the coffee 200°f

$200 = ae^{0} + 75$

$200 - 190 = 010$

$f(2) = ae^{-kt} + c$

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: Fang, Suyi WA3005435

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a = 27.03$ $\quad b = 1.67$ $\quad r = 0.67$

$1.67 = 1 + r$

$.67 = r$

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

$a = 12.75$

$r = 4.1\% = 0.041$

$p(t) = 12.75(1+0.041)^t$

$p(t) = 12.75(1.041)^t$

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

$P(t) = 12.75(1.041)^t$

$P(25) = 12.75(1.041)^{25}$

$\approx \$34.82$

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

$P(t) = a + mt$

$P(t) = 6500 - 60t$

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

$P(t) = a(1-r)^t$

$P(t) = 6500(1-0.05)^t$

$P(t) = 6500(0.95)^t$

4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

$y = 0$

5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$ 0

$192(0.75)^5 = 45.56$

$192(0.75)^9 = 14.42$

$192(0.75)^{15} = 2.56$

$192(0.75)^{35} = 0.0081$

The long term behavior is that it will get closer and closer to 0 as the x goes up.

(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$ 0

$2(3)^{-2} = 0.223$

$2(3)^{-8} = 0.00030$

$2(3)^{-15} = 0.000000139$

The longterm behavior that it will get closer and closer to 0

(c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to$ 15

$(0.9)^t = 0$

As 0.9 to a bigger and bigger power, it gets to 0.

$5(3+0) = 15$

$5(3 + (0.9)^6) = 17.657$

$5(3 + (0.9)^{60}) = 15.0089$

$5(3 + (0.9)^{400}) = 15$

$5(3 + (0.9)^{405}) = 15$

(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$ 0

$8e^{0.5(5)} = 0.65668$

$8e^{0.5(15)} = 0.00442$

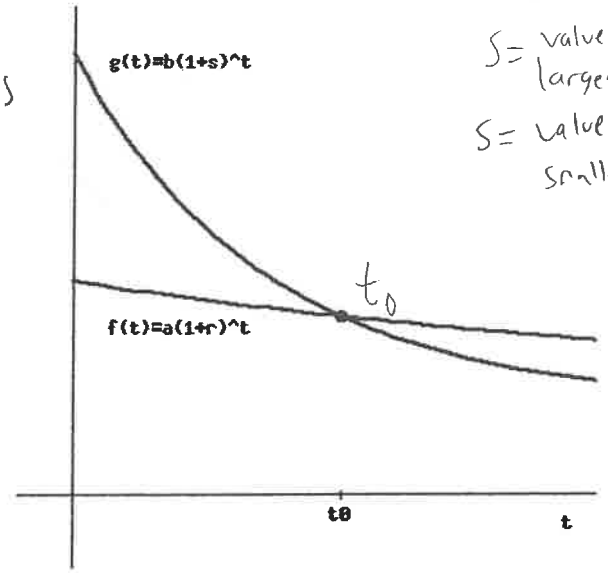
$8e^{0.5(27)} = 0.000011$

get closer to 0 as t goes smaller

6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.

$g(t) = b(1+s)^t$

t_0 decreases



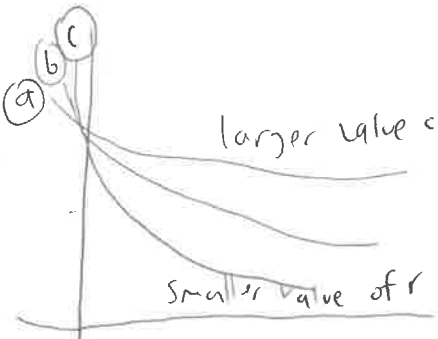
s = value decreased
larger r

s = value increased
smaller r

⑨ $g(t) = 10(1+($
$= 10(0.$

⑥ $g(t) = 10(1+($
$= 10(0.5$

⑤ $g(t) = 10(1+(-0.$
$= 10(0.1)^t$



7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$m(0) = 160\,e^{-0.025(0)}$
$= 160e^0$
$= 160$ grams

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

$m(10) = 160\,e^{-0.025t}$
$= 160\,e^{-0.025(10)}$
$= 160\,e^{-0.25}$ or ≈ 124.6081

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

   (a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order. to the left
   shift the x value -4, then shift the graph down up 2 units

   (b) (4 pts) State the domain and range of $f(x)$.
   domain: $(-\infty, \infty)$
   range $(-4, \infty)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

   (a) (4 pts) $\ln 10 = y$
   $e^y = 10$

   (b) (4 pts) $\ln x = 5$
   $e^5 = x$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

   (a) (4 pts) $\ln(e^{1.7}) = 1.7$

   (b) (4 pts) $e^{\ln 4} = 4$

   (c) (4 pts) $\ln\left(\frac{1}{e^3}\right) = \ln(e^{-3}) = -3$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

$$e^{-3x+2} = 10$$
$$\ln(e^{-3x+2}) = \ln 10$$
$$-3x+2 = \ln 10$$
$$\left(-\tfrac{1}{3}\right) -3x = -2 + \ln 10 \left(-\tfrac{1}{3}\right)$$
$$\boxed{x = -\frac{\ln 10 - 2}{3}} \approx -0.1009$$

(b) (4 pts) $8 - \ln(2-x) = 1$.

$$8 - \ln(2-x) = 1$$
$$(-1) -\ln(2-x) = -7 \ (-1)$$
$$\ln(2-x) = 7$$
$$e(\ln(2-x)) = e^7$$
$$2-x = e^7$$
$$(-1) -x = e^7 - 2 \ (-1)$$
$$\boxed{x = -e^7 + 2} \approx 2.0009$$

(c) (4 pts) $4^x = 3$

$$\ln(4^x) = \ln 3$$
$$x \ln 4 = \ln 3$$
$$\boxed{x = \frac{\ln 3}{\ln 4}} \approx 0.7925$$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$$2 \cdot 4^{2x-4} = 60$$
$$4^{2x-4} = 30$$
$$\ln(4^{2x-4}) = \ln 30$$
$$2x-4 \ln 4 = \ln 30$$
$$2x-4 = \frac{\ln 30}{\ln 4}$$
$$\left(\tfrac{1}{2}\right) 2x = \frac{\ln 30}{\ln 4} + 4 \left(\tfrac{1}{2}\right)$$
$$\boxed{x = \frac{\ln 30}{2 \ln 4} + 2} \approx 4.3575$$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

$a = 125$ $\quad c = 75$

$200 = ae^{-k(0)} + 75$

$200 = a + 75$

$125 = a$

$190 = 125e^{-2k} + 75$

$115 = 125e^{-2k}$

$\frac{115}{125} = e^{-2k}$

$\ln\left(\frac{23}{25}\right) = \ln e^{-2k}$

$\ln\left(\frac{23}{25}\right) = -2k$

$\frac{-\ln\left(\frac{23}{25}\right)}{2} = k$

$0.04169 \approx k$

$F(t) = 125e^{-0.04169t} + 75$

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

***

Last name, first name, ID#: Bornoski, Elisabeth, WA3279774

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a = 27.03$ $\quad b = 1.67$ $\quad r = 0.67 = 67\%$? check percent

$a = 27.03$
$b = 1.67$
$c =$

growth rate : $r = 1.67 - 1 = 0.67 = 67\%$

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

starting price : \$12.75, increasing 4.1% annually

$P(t) = a(1+r)^t$

$a = \$12.75$
$r = 0.041$

$P(t) = 12.75(1.041)^t$

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

After 25 years,
$t = 25$

$P(25) = 12.75(1.041)^{25}$
$=$ approximately \$34.82

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

starting population : 6500
decreases by 60 people per year,

$P(t) = P_0 + mt$

$P(t) = 6500 - 60t$

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t. decreases by 5% annually

$P(t) = a(1+r)^t$
$r = -0.05$
$P(t) = 6500(1-0.05)^t$

$P(t) = 6500(0.95)^t$

4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

$y = ab^+$, where $0 < b < 1$

as $+ \longrightarrow$

5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$ [0]

$192(0.75)^x$

greater = > than

less = < than

(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$ [0]

$2(3)^+$

(c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to$ [19.5]

$5(3+(0.9)^+)$

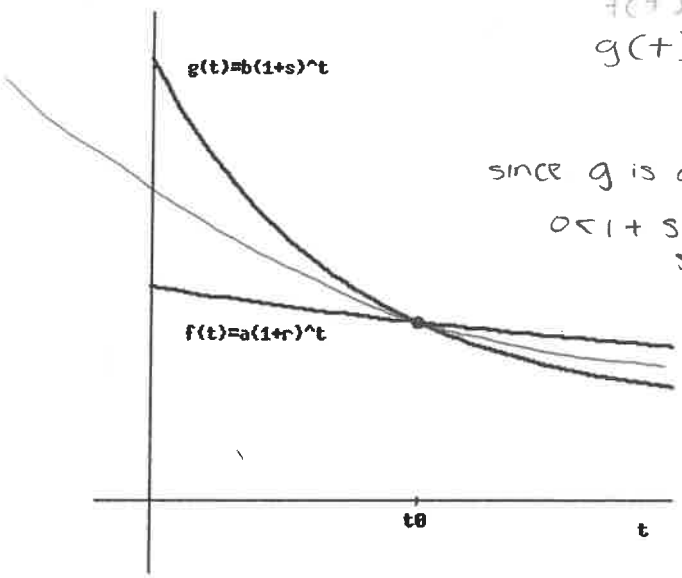
(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$ [0]

$8e^{0.5+}$

6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.

$f(t) = a(1+r)^t$

$g(t) = b(1+s)^t$



since g is decreasing,

$0 < 1 + s < 1$,

so $s < 1$

t_0 decreases

7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$m(0) = 160e^{-0.025(0)}$

$= 160e^0$

$e^0 = 1$

$= 160$ grams

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

substitute $t = 10$ into the original formula

$m(10) = 160e^{-0.025(10)}$

$= 160e^{-0.25}$

$= 140$ grams

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

(a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

Given = $g(x) = e^x$

$f(x) = e^{x-4} + 2$

e^{-x}

$e^{-x} - 4$ The order does not matter

(b) (4 pts) State the domain and range of $f(x)$.

Domain = $(-\infty, \infty)$

Range = $(-4, \infty)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$. $\ln A = B \iff e^B = A$

(a) (4 pts) $\ln 10 = y$

$e^y = 10$

(b) (4 pts) $\ln x = 5$

$e^5 = x$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

(a) (4 pts) $\ln(e^{1.7})$

$\ln(e^x) = x$ $= 1.7$

(b) (4 pts) $e^{\ln 4}$

$e^{\ln x} = x$ $= 4$

(c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

$\frac{1}{e^3} = e^{-3}$

$\ln(e^{-3}) = -3$

$= -3$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

$$e^{-3x+2} = 10$$

$$\ln(e^{-3x+2}) = \ln 10$$

$$-3x + 2 = \ln 10$$

$$-3x = \ln 10 - 2$$

$$\boxed{x = \frac{\ln 10 - 2}{-3}}$$

(b) (4 pts) $8 - \ln(2 - x) = 1$.

$$8 - \ln(2-x) = 1$$

$$-\ln(2-x) = 1 - 8 = -7$$

$$\ln(2-x) = 7$$

$$2 - x = e^7$$

$$-x = e^7 - 2$$

$$\boxed{x = 2 - e^7}$$

(c) (4 pts) $4^x = 3$

$$\ln(4^x) = \ln 3$$

$$x \ln 4 = \ln 3$$

$$\boxed{x = \frac{\ln 3}{\ln 4}}$$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$$2 \cdot 4^{2x-4} = 60$$

$$4^{2x-4} = \frac{60}{2}$$

$$(2x-4)\ln 4 = \ln\left(\frac{60}{2}\right)$$

$$2x - 4 = \frac{\ln(60/2)}{\ln 4}$$

$$\boxed{x = 4 + \frac{\ln(60/2)}{2\ln 4}}$$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

initial coffee temp : 200°F
room temp : 75° F
temp after 2 minutes : 190°F

$F(t) = ae^{-Kt} + C$

room temp = horizontal asymptote

$C = 75$

At $t = 0$

$200 = ae^{0} + 75$

$200 = a + 78$

$a = 122$

$F(t) = 122e^{-Kt} + 75$

$t = 2$ & $F(2) = 190$

$190 = 122e^{-2K} + 75$

$115 = 122e^{-2K}$

$\ln\left(\frac{115}{122}\right) = -2K$

$K = -\frac{\ln(115/122)}{2}$

$K \approx 0.0295$

$$F(t) = 122e^{-0.0295t} + 75$$

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: Urquhart Anne WA32675900

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a =$ 27.03 $\quad b =$ 1.67 $\quad r =$ 0.67

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

   $P(t) = 12.75\ (1 + 0.041)^t$

   $\boxed{P(t) = 12.75\ (1.041)^t}$

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

   $P(25) = 12.75\ (1.041)^{25}$

   $\boxed{P(25) = 34.82}$

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

   $\boxed{P(t) = 6500 - 60t}$

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

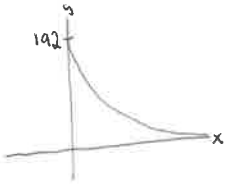
   $\boxed{P(t) = 6500\ (0.95)^t}$

4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

$y = 0$

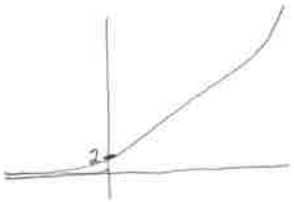
5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$ 0

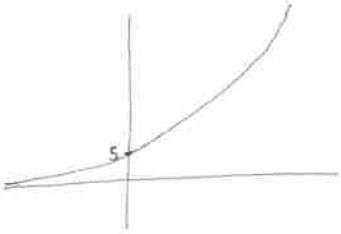


$f(10) = 192(0.75)^{10}$
$f(10) = 10.81$
$f(100) = 6.16$
$f(500) = 0.000109$
$f(1000) = 0$ (...)

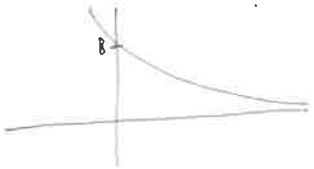
(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$ 0



(c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to \infty$

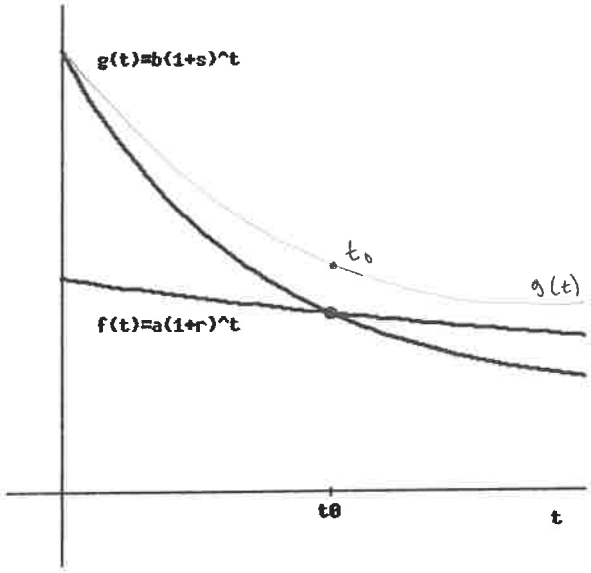


(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$ ∞



6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.

the value of t_0 increases



7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$m(0) = 160e^{-0.025(0)}$

$m(0) = 160e^{0}$

$m(0) = 160 \cdot 1$

$\boxed{m(0) = 160}$

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

$m(10) = 160e^{-0.025(10)}$

$m(10) = 160e^{-0.25}$

$m(10) \approx 124.6081$

$\boxed{m(10) \approx 124.6081}$

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

   (a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

   horizontal shift by 4 units and vertical shift by two units. The order does not matter in this case.

   (b) (4 pts) State the domain and range of $f(x)$.

   dom $f(x) = (-\infty, \infty)$

   ran $f(x) = (2, \infty)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

   (a) (4 pts) $\ln 10 = y$

   $e^y = 10$

   (b) (4 pts) $\ln x = 5$

   $e^5 = x$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

    (a) (4 pts) $\ln(e^{1.7})$

    $\ln(e^{1.7}) = 1.7$

    (b) (4 pts) $e^{\ln 4}$

    $e^{\ln 4} = 4$

    (c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

    $\ln\left(\frac{1}{e^3}\right) = -3$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

$\ln\left(e^{-3x+2}\right) = \ln(10)$

$-3x+2 = \ln(10)$

$-3x = \ln(10) - 2$

$x = \frac{\ln(10)-2}{-3}$

$x+2 = \frac{\ln(10)}{-3}$

$\boxed{x = \frac{\ln(10)}{-3} - 2}$

(b) (4 pts) $8 - \ln(2-x) = 1$.

$8 - \ln(2-x) = 1$

$-\ln(2-x) = -7$

$\ln(2-x) = 7$

$e^7 = 2-x$

$e^7 - 2 = -x$

$\boxed{-e^7 + 2 = x}$

(c) (4 pts) $4^x = 3$

$\ln(4^x) = \ln(3)$

$x \cdot \ln(4) = \ln(3)$

$\boxed{x = \frac{\ln(3)}{\ln(4)}}$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$4^{2x-4} = 30$

$\ln\left(4^{2x-4}\right) = \ln(30)$

$(2x-4)\cdot\ln(4) = \ln(30)$

$2x-4 = \frac{\ln(30)}{\ln(4)}$

$x-4 = \left(\frac{\ln(30)}{\ln(4)}\right) \div 2$

$\boxed{x = \left(\left(\frac{\ln(30)}{\ln(4)}\right) \div 2\right) + 4}$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

$$F(t) = 125e^{\left(\frac{\ln\left(\frac{115}{125}\right)}{2}\right)(t)} + 75$$

$F(2) = 190 \qquad c = 75 \qquad a = 125$

$$F(2) = 125e^{-k(2)} + 75$$

$$190 = 125e^{-k(2)} + 75$$

$$115 = 125e^{-k(2)}$$

$$\frac{115}{125} = e^{-k(2)}$$

$$\ln\left(\frac{115}{125}\right) = \ln\left(e^{-k(2)}\right)$$

$$\ln\left(\frac{115}{125}\right) = -k \cdot 2$$

$$\frac{\ln\left(\frac{115}{125}\right)}{2} = -k$$

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

***

Last name, first name, ID#: Engalan, Stephanie Nicole, S00017399

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a = 27.03$ $b = 1.67$ $r = 0.67$

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

$P(t) = ab^t$
$a = 12.75$
$= 0.041$

(a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

$P(t) = a(1+r)^t$
$= 12.75(1+0.041)^t$

(b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

$P(t) = a(1+r)^t$
$P(t) = 12.75(1+0.041)^{25}$
$= 34.82$

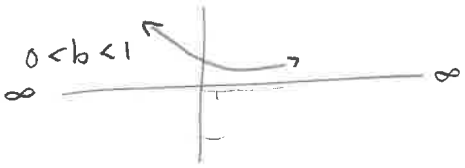
3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

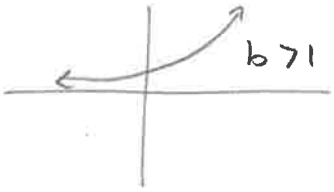
(a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

$P(t) = y_0 + m(t - t_0)$
$= 6500 - 60t$

(b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

$P(t) = ab^t$
$P(t) = 6500(0.05)^t$





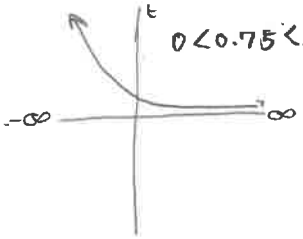
4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

yes, there is a horizontal asymptote

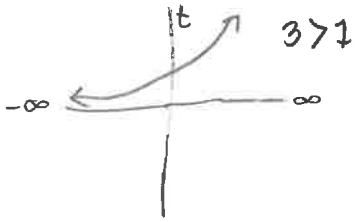
$y = 0$

5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$ 0



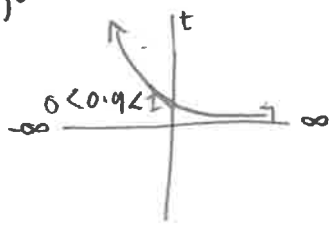
(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$ 0



(c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to$ 10

$f(t) = 5(3+(0.9))^t$

$=$



$f(t) = 5(2+(0.9))t$

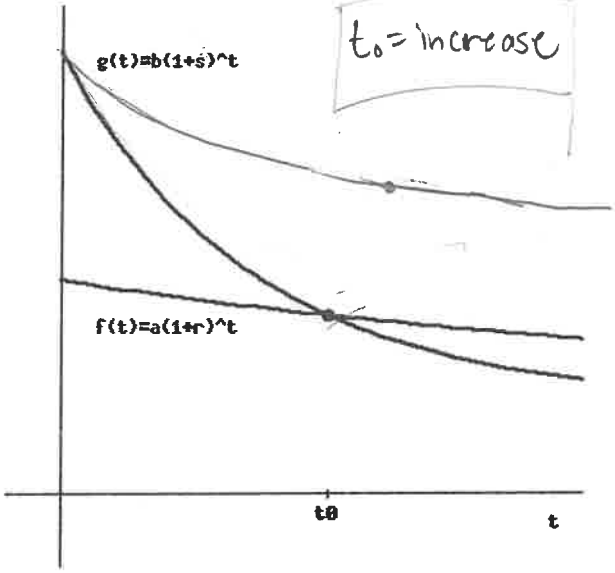
$= 5(2+0)t$

$= 5(2)$

$= 10$

(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$ 0

6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.



t_0 = increase

7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$m(t) = 160e^{-0.025t}$

$m(0) = 160e^{-0.025(0)}$

$= 160e^{0}$

$= 160(1)$

$= 160$ grams

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

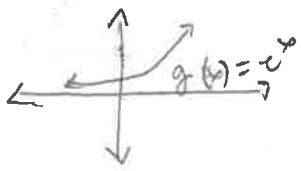
$m(10) = 160e^{-0.025(10)}$

$= 160e^{-0.25}$

$= 124.61$ grams

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

   (a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.



vertical shift up by 2 units
scaling by -4

   (b) (4 pts) State the domain and range of $f(x)$.

Domain = $(-\infty, +\infty)$

Range = $(2, \infty)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

   (a) (4 pts) $\ln 10 = y$

$e^y = 10$

   (b) (4 pts) $\ln x = 5$

$e^5 = x$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

   (a) (4 pts) $\ln(e^{1.7})$ $= 1.7$

   (b) (4 pts) $e^{\ln 4}$ $= 4$

   (c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

$= \ln(e^{-3})$

$= -3$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

$e^{-3x+2} = 10$

$\ln(e^{-3x+2}) = \ln(10)$

$-3x+2 = \ln(10) - 2$

$-2 \quad\quad -2$

$\frac{-3x}{-3} = \frac{\ln(10)-2}{-3}$

$\boxed{x = \frac{\ln(10)-2}{-3}}$

(b) (4 pts) $8 - \ln(2-x) = 1$.

$8 - \ln(2-x) = 1 - 8$

$-8 \quad\quad -8$

$\frac{-\ln(2-x)}{-1} = \frac{-7}{-1}$

$\ln(2-x) = \frac{-7}{-1} \rightarrow -7 = 7$

$e^{\ln(2-x)} = e^7$

$2 - x = e^7 - 2$

$-2 \quad\quad -2$

$\frac{-x}{-1} = \frac{e^7-2}{-1}$

$x = \frac{e^7-2}{-1}$

$\boxed{x = -e^7 + 2}$

(c) (4 pts) $4^x = 3$

$4^x = 3$

$\ln(4)^x = \ln(3)$

$x \ln(4) = \ln(3)$

$\frac{x \ln(4)}{\ln(4)} = \frac{\ln(3)}{\ln(4)}$

$\boxed{x = \frac{\ln(3)}{\ln(4)}}$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$\frac{2 \cdot 4^{2x-4}}{2} = \frac{60}{2}$

$4^{2x-4} = 30$

$\ln(4)^{2x-4} = \ln(30)$

$\frac{(2x-4)\ln(4)}{\ln(4)} = \frac{\ln(30)}{\ln(4)}$

$2x - 4 = \frac{\ln(30)}{\ln(4)}$

$+4 \quad\quad +4$

$2x = \frac{\ln(30)}{\ln(4)} + 4$

$\frac{1}{2} \cdot 2x = \frac{1}{2}\left[\frac{\ln(30)}{\ln(4)} + 4\right]$

$x = \frac{1}{2} \cdot \frac{\ln(30)}{\ln(4)} + \frac{1}{2} \cdot \frac{4}{1}$

$\boxed{x = \frac{\ln(30)}{\ln(4)} + 1}$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

Given:

$t=0, P(0)=200°F$

$t=2, P(2)=190°F$

$t\to\infty, P(t)=75°F$

$t\to\infty, P(t)\to C$

$C = 75°F$

$F(0) = a + C$

$200 = a + 75$

$-75 \quad -75$

$a = 125$

$F(2) = ae^{-kt} + C$

$190 = 125e^{-k(2)} + 75$

$-75 \quad -75$

$\frac{115}{125} = \frac{125e^{-k(2)}}{125}$

$\frac{115}{125} = e^{-k(2)}$

$\ln\left(\frac{115}{125}\right) = \ln\left(e^{-k(2)}\right)$

$\frac{\ln\left(\frac{115}{125}\right)}{-2} = \frac{-k(2)}{-2}$

$k = \frac{\ln\left(\frac{115}{125}\right)}{-2}$

$k \approx 0.0417$

$F(t) = ae^{-kt} + C$

$F(t) = 125e^{-0.0417t} + 75$

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: de Haro, Alex

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a = 27.03$ $b = 1.67$ $r = .67$

$1+r = 1.67$

$r = .67$

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

$a = 12.75$

$b = 1.041$

$r = 0.041$

$1 + r = b$

$P(t) = 12.75(1.041)^t$

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

$P(25) = 12.75(1.041)^{25} = \$34.82$

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

$P(t) = 6500 - 60t$

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

$1 - 0.05 = b$

$0.95 = b$

$P(t) = 6500(0.95)^t$

4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

$b > 0$

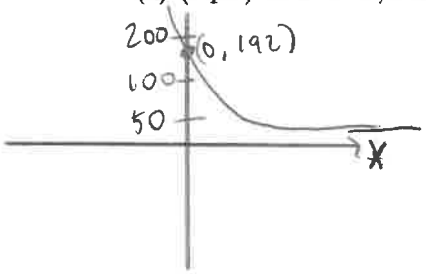
$\boxed{y = 0}$

the horizontal asymptote is 0

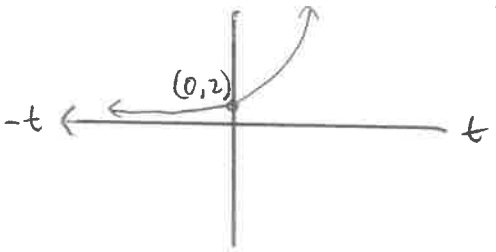
we assume $c = 0$

5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$ $\boxed{0}$



(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$ $\boxed{0}$

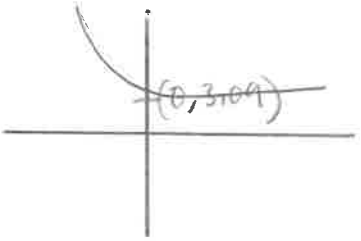


ex. $2(3)^{-1} = 0.66$, $(-1, 0.66)$

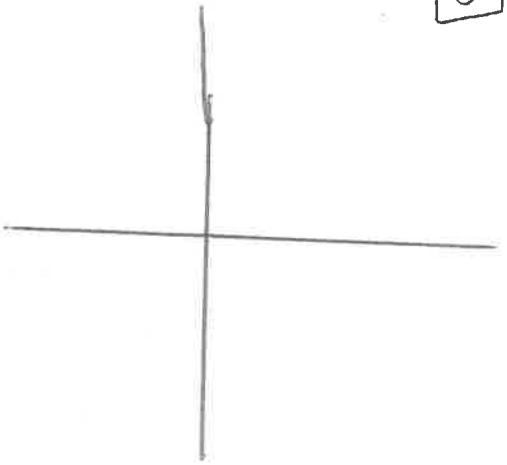
$2(3)^{-2} = 0.22$, $(-2, 0.22)$

moving towards zero as $t \to -\infty$.

(c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to$ $\boxed{3.09}$



(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$ $\boxed{0}$

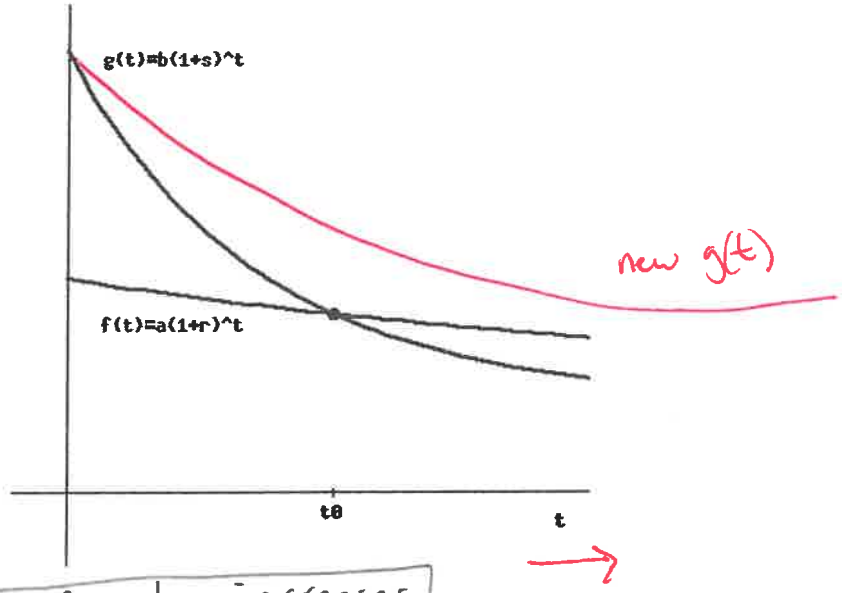


$8e^{0.5(-1)} = 4.85$

$8e^{0.5(-4)} = 1.082$

as $t \to \infty$, the output gets closer to zero

6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.



the value of t_0 increases because the two lines would intersect farther down the x axis, towards infinity.

7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$m(0) = 160e^{-0.025(0)} \Rightarrow$ $m(0) = 160$ g

$(e^0 = 1)$

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

$m(10) = 160e^{-0.025(10)}$

$m(10) \Rightarrow 160e^{-0.25}$

$m(10) \Rightarrow 160(0.7788)$

after 10 years, $\approx$ 124.6081 grams of the mass remains

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

(a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

First, shift the graph of $g(x)$ to the right by 4 & then shift up by a factor of 2.

(b) (4 pts) State the domain and range of $f(x)$.

DOM: $(-\infty, \infty)$

RAN: $(-\infty, 2)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

(a) (4 pts) $\ln 10 = y$

$e^y = 10$

(b) (4 pts) $\ln x = 5$

$e^5 = x$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

(a) (4 pts) $\ln(e^{1.7})$

1.7

(b) (4 pts) $e^{\ln 4}$

4

(c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

$\ln(e^{-3}) = -3$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

$e^{-3x+2} = 10 \Rightarrow -3x+2 \approx 2.302585$

$(\ln)\quad (\ln)$

$-2 \quad -2$

$\frac{-3x}{-3} \approx \frac{0.302585}{-3}$

$x \approx -0.10086$

(b) (4 pts) $8 - \ln(2-x) = 1$.

$8 - \ln(2-x) = 1 \Rightarrow \ln(2-x) = 7$

$-8 \quad -8 \qquad (e) \quad (e)$

$\Rightarrow 2 - x \approx 1096.633 \Rightarrow x \approx -1094.633$

$-2 \quad -2$

(c) (4 pts) $4^x = 3$

$4^x = 3 \Rightarrow x \cdot \ln(4) = \ln(3)$

$(\ln)\ (\ln)$

$\frac{x \cdot 1.38629}{1.38629} = \frac{1.0986}{1.38629}$

$x \approx 0.7924$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$\frac{2 \cdot 4^{2x-4}}{2} = \frac{60}{2}$

$4^{2x-4} = 30$

$(\ln)\quad (\ln)$

$(2x-4)\frac{(\ln 4)}{\ln(4)} = \frac{\ln(30)}{\ln(4)}$

$2x - 4 = \frac{\ln(30)}{\ln(4)}$

$+4 \qquad +4$

$2x = \frac{\ln(30)}{\ln(4)} + 4$

$x \approx 3.226$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

$F(0) = 200$

$F(2) = 190$

$a = 125$

$k = 0.04169$

$200 = ae^{-k(0)} + 75$

$200 = a + 75$

$-75 \qquad -75$

$\boxed{a = 125}$

$190 = 125e^{-k(2)} + 75$

$-75 \qquad -75$

$\frac{115}{125} = \frac{125e^{-k(2)}}{125}$

$\frac{115}{125} = e^{-k(2)}$

$(\ln) \qquad (\ln)$

$\frac{\ln\left(\frac{115}{125}\right)}{2} = \frac{-k(2)}{2}$

$\frac{-0.04169}{-1} = \frac{-k}{-1}$

$k \approx 0.04169$

$\boxed{F(t) = 125e^{-0.04169(t)} + 75}$

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: Luna, Penelope WA3000603

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a = 27.03$ $b = 1.67$ $r = .67$ or 67%.

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

   $a = 12.75$ $r = 4.1$

   $P(t) = 12.75(1.041)^t$

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

   $P(30) = 12.75(1.041)^{25}$

   a movie ticket will cost approximately \$34.82 dollars in 25 years

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

   $P(t) = 6500 - 60t$

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

   $P(t) = 6500(.95)^t$

   $1 + (-0.05) = .95$

4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

no horizontal asymptote ?

not too sure tbh :)

5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to 0$

$0 < 0.75 < 1$

(b) (4 pts) As $t \to -\infty$, $2(3)^t \to 0$

$b = 3 > 1$

(c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to 15$

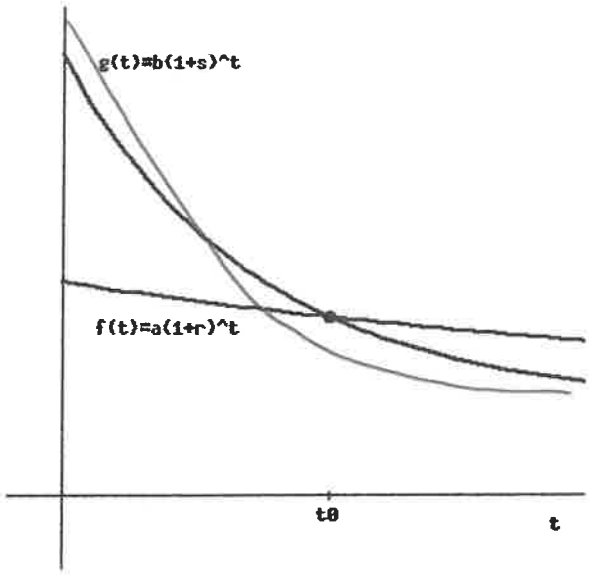
$1 < 0.9 < 1$

$5(3+0) = 15$

(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to 0$

$0.5t \to -\infty$, $e^{0.5} \to 0 \times 8 = 0$

6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.



t_0 increases because first graph stays untouched while the other graph increases the intersection time becomes bigger

7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$m(0) = 160e^{-0.025(0)} = 0$

$160e^0 \rightarrow 160(1)$

$m(0) = 160$

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

$-0.025(10) = -0.25$

$160e^{-0.25}$ $\quad e^{-0.25} \approx 0.7788 \times 160 =$ 124.608 mass rem after 10 y

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

   (a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

   reflects across the x axis, shifts UP 2 units

   (b) (4 pts) State the domain and range of $f(x)$.

   Domain: $(-\infty, \infty)$

   Range: $(2, \infty)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

   (a) (4 pts) $\ln 10 = y$

   $e^y = 10$

   (b) (4 pts) $\ln x = 5$

   $e^5 = x$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

    (a) (4 pts) $\ln(e^{1.7})$

    1.7

    (b) (4 pts) $e^{\ln 4}$

    4

    (c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

    -3

    $\frac{1}{e3} = e^{-3}$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

$$\ln\left(e^{-3x+2}\right) = \ln(10)$$

$$-3x+2 = \ln(10)$$
$$-2 \qquad -2$$

$$\frac{-3x}{-3} = \frac{\ln(10)-2}{-3}$$

$$\boxed{x = \frac{\ln(10)-2}{-3}}$$

(b) (4 pts) $8 - \ln(2-x) = 1$.

$$-8 \qquad -8$$

$$-1 \cdot -\ln(2-x) = -7 \cdot -1$$

$$\ln(2-x) = 7$$

$$e^7 = 2-x$$
$$-2 \quad -2 \quad \cdot -1$$

$$\boxed{x = 2-e^7}$$

(c) (4 pts) $4^x = 3$

$$\frac{x\ln(4)}{4} = \frac{\ln(3)}{4}$$

$$\boxed{x = \frac{\ln(3)}{4}}$$

(d) (4 pts) $\frac{2 \cdot 4^{2x-4}}{2} = \frac{60}{2}$

$$\ln(4^{2x-4}) = \ln(30)$$

$$(2x-4)\ln(4) = \ln(30)$$
$$+4$$

$$\boxed{x = 11 + \frac{\ln(30)}{2\ln(4)}}$$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

2min 200F 190F 75F

$200 = ae^0 + 75$

$-75 \qquad -75$

$a = 125$

$F(t) = 125e^{-kt} + 75$

$F(2) = 190$

$190 = 125e^{-2k} + 75$

$-75 \qquad -75$

$\frac{115}{125} = \frac{125e^{-2k}}{125}$

$\frac{\ln(0.92)}{-2} = \frac{-2k}{-2} \rightarrow k = \frac{-\ln(0.92)}{2} \rightarrow k \approx 0.7765$

$F(t) = 125e^{-0.7765t} + 75$

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: Huang, David WA3278621

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a = 27.03$ $b = 1.67$ $r = 0.67$

$1+r = 1.67$
$-1 \quad -1$
$r = 0.67$

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

(a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

$a = 12.75$
$b = 0.041 + 1 = 1.041$
$P(30) = 12.75(1.041)^{30}$
≈ 42.56

The price would be \$42.56 30 years from today.

(b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

$P(25) = 12.75(1.041)^{25}$
$\approx \$34.81$

It would cost \$34.81 25 years from today.

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

(a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

t-int: 6500
Slope: -60

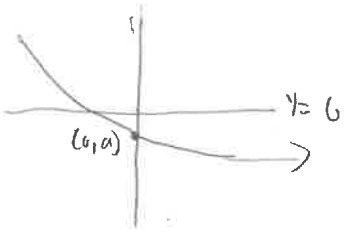
(b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

5% decrease $= -0.05$

$1+r = 0.95$

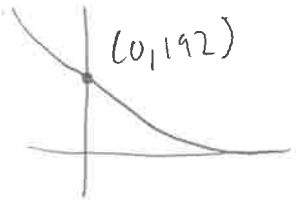
$P(t) = 6500 - 60t$
$P(t) = ab^t = 6500(0.95)^t$

4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

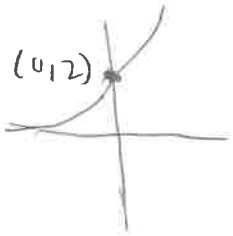


5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

   (a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$

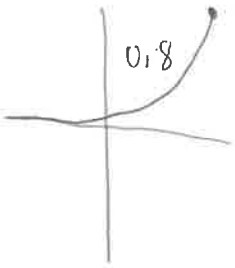


   (b) (4 pts) As $t \to -\infty$, $2(3)^t \to$

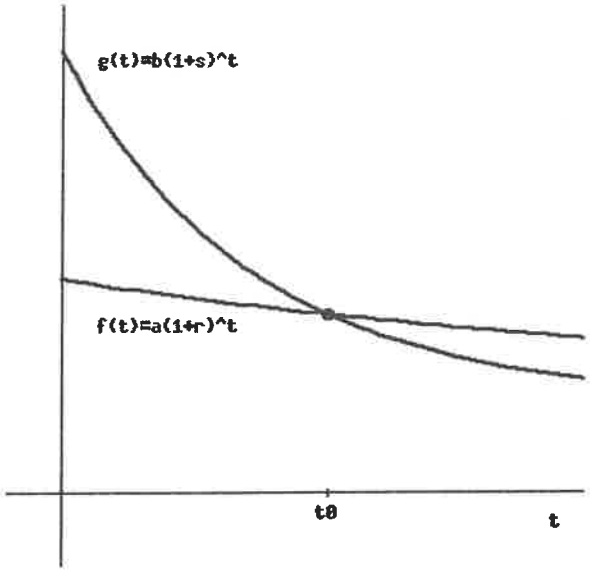


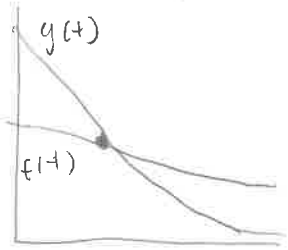
   (c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to$

   (d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$



6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.





It would decrease if value of s is increased.

7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$m(0) = 160e^{-0.025 \cdot 0} = 160e^{0} = 160$ grams

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

$m(10)$

$160e^{-0.025 \cdot 10}$

$= 160e^{-0.25}$ $\quad e^{-0.25} \simeq 0.7788$

160×0.7788

$\simeq 124.6$

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

   (a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

   X means reflect across the x axis
   -4 means shift down 4
   +2 means shift up 2

   (b) (4 pts) State the domain and range of $f(x)$.

   D: $(-\infty, \infty)$
   R: $(2, \infty)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

   (a) (4 pts) $\ln 10 = y$

   $e^y = 10$

   (b) (4 pts) $\ln x = 5$

   $e^5 = x$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

    (a) (4 pts) $\ln(e^{1.7})$ $e \ln e^{1.7} = 1.7$

    (b) (4 pts) $e^{\ln 4}$ $e \ln e^4 = 4$

    (c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

    fraction flips to negative exponent

    $e \ln e^{-3} = \ln -3$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

$e \simeq 2.718$

$\log e^{-3x+2} = \log 10$

$-3x+2 \log e = \log 10$

$x = -0.1009$

$\frac{-3x+2 \log e}{\log e} = \frac{\log 10}{\log e}$

$-3x+2 = 2.3028$

$-2 \quad -2$

$\frac{-3x}{-3} = \frac{0.3028}{-3}$

(b) (4 pts) $8 - \ln(2-x) = 1$.

$8 - \ln(2-x) = 1$

$\ln e(2-x) = 9$

$e^9 = 2-x$

$x = 2 - e^9$

(c) (4 pts) $4^x = 3$

$4x = 3$

$\frac{\log 3}{\log 4}$

$x = 0.7924$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$\frac{2 \cdot 4^{2x-4}}{2} = \frac{60}{2}$

$4^{2x-4} = 30$

$\log 4^{2x-4} = \log 30$

$2x-4 \frac{\log 4}{\log 4} = \frac{\log 30}{\log 4}$

$2x-4 = 2.4534$

$+4 \quad +4$

$\frac{2x}{2} = \frac{6.4534}{2}$

$\simeq 3.2267$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

$c = 75$

$a = 125$

$k = 0.0416$

$200 = a + 75$

So, $a = 125$ and $c = 75$

$f(2) = 190$ and $f(2) = ae^{-k(2)} + c$

$190 = 125e^{-k2} + 75$

$190 - 75 = 125e^{-k2}$

$\frac{115}{125} = \frac{125e^{-k2}}{125}$

$\frac{115}{125} = e^{-k2}$

$\frac{\ln\frac{115}{125}}{-2} = \frac{\ln e^{-k2}}{-2}$

$k = \frac{\ln\frac{115}{125}}{-2} \approx 0.0416$

$F(t) = 125e^{-0.0416\left(\ln\left(\frac{115}{125}\right)\right)t} + 75$

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: Marsy Rose W0318813

Please read the following instructions carefully.

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- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a = 27.03$ $\quad b = 1.67$ $\quad r = 0.67$

$b = 1+r$

$r = b-1$

$= 1.67 - 1$

$r = 0.67$

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation. $P(t) = a(1+r)^t$

Starting value $a = 12.75$

growth rate $k = 4.1\% = 4.1 \times \frac{1}{100} = 0.041$

growth Factor $b = 1+r$

$= 1+0.041$

$= 1.041$

$P(t) = a(1+r)^t$

$= 12.75(1.041)^t \Rightarrow$ a formula for the cost of a movie ticket t for years

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

a movie cost in 25 years, when $t = 25$

$P(t) = a(1+r)^t$

$P(25) = 12.75(1.041)^{25}$

$P(25) \approx 34.81$ tickets

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

$a = 6500$, $b = -60$

Linear function,

$P(t) = a+bt$

$= 6500 + (-60)t$

$= 6500 - 60t$

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

If the population decreases by 5% per year,

$a = 6500$

$r = -5\%$

$= -5 \times \frac{1}{100}$

$r = -0.05$

$P(t) = a(1+r)^t$

$= 6500(1+(-0.05))^t$

$= 6500(1-0.05)^t$

$P(t) = 6500(0.95)^t$

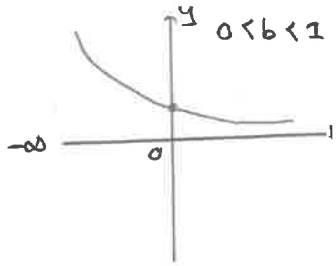
4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

$y = ab^t$ as $t \to +\infty$

$ab^t \to a(0) = 0$

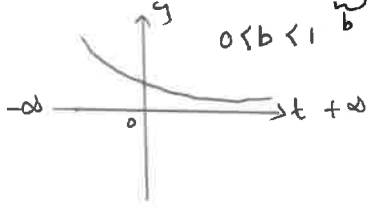
$\therefore$ so $y = 0$

Yes, the horizontal asymptote is $y = 0$

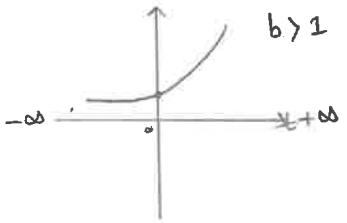


5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

(a) (4 pts) As $x \to \infty$, $192(\underbrace{0.75}_{b})^x \to$ [0]

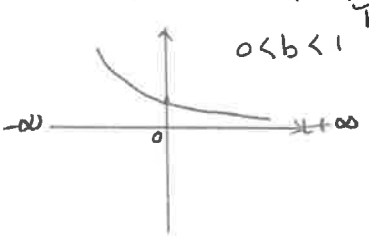


(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$ [0]

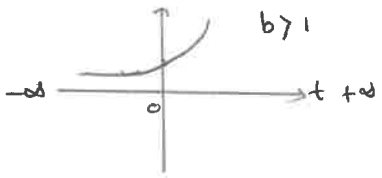


(c) (4 pts) As $t \to \infty$, $5(3 + (\underbrace{0.9}_{b})^t) \to$ [15]

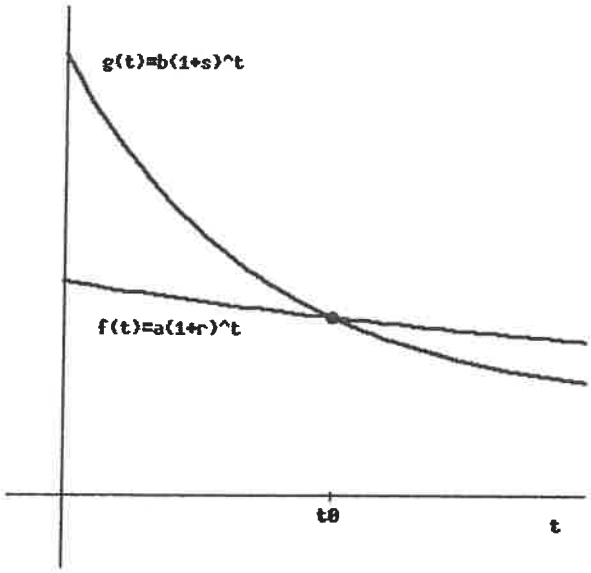
5(3+0)



(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$ [0]



6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.



The value of t_0 decreases because increasing makes decrease more slowly, so the curves intersect earlier.

7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$m(0) = 160e^{-0.025(0)}$

$= 160e^0 \quad (\because e^0 = 1)$

$m(0) = 160$ grams

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

The mass remains after 10 years, $t = 10$

$m(10) = 160e^{-0.025(10)}$

$m(10) \approx 124.61$ grams

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

   (a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

   Scaling by $+1x - 4$ reflection over the y axis.

   Vertical shift upward to 2 units.

   (b) (4 pts) State the domain and range of $f(x)$.

   Domain of $f(x)$: $(-\infty, \infty)$

   Range of $f(x)$: $(+2, \infty)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

   (a) (4 pts) $\ln 10 = y$

   $\log_e 10 = y$

   $e^y = 10$

   (b) (4 pts) $\ln x = 5$

   $\ln x = 5$

   $e^5 = x$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

    (a) (4 pts) $\ln(e^{1.7})$

    $\ln(e^{1.7}) = 1.7$

    (b) (4 pts) $e^{\ln 4}$

    $e^{\ln 4} = 4$

    (c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

    $\ln\left(\frac{1}{e^3}\right) = -3$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

$e^{-3x+2} = 10$

$\ln(e)^{-3x+2} = \ln(10)$

$-3x+2 = \ln(10)$

$-3x = \ln(10) - 2$

$x = -\frac{\ln(10)-2}{3}$ or $-\frac{\ln(10)}{3} + \frac{2}{3}$

(b) (4 pts) $8 - \ln(2-x) = 1$.

$8 - \ln(2-x) = 1$

$-\ln(2-x) = 1-8$

$+\ln(2-x) = +7$

$\ln(2-x) = 7$

$e^{\ln(2-x)} = e^7$

$2-x = e^7$

$-x = e^7 - 2 \Rightarrow$

$\Rightarrow \frac{-x}{-1} = \frac{e^7-2}{-1}$

$\boxed{x = -e^7 + 2}$

(c) (4 pts) $4^x = 3$

$\ln(4)^x = \ln(3)$

$x\ln(4) = \ln(3)$

$x = \frac{\ln(3)}{\ln(4)}$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$2 \cdot 4^{2x-4} = 60$

$4^{2x-4} = \frac{60}{2}^{30}$

$4^{2x-4} = 30$

$\ln(4)^{2x-4} = \ln(30)$

$2x-4\ln(4) = \ln(30)$

$2x-4 = \frac{\ln(30)}{\ln(4)}$

$2x = \frac{\ln(30)}{\ln(4)} + 4$

$x = \frac{1}{2}\left[\frac{\ln(30)}{\ln(4)} + 4\right] \Rightarrow$

$\Rightarrow \frac{\ln(30)}{2\ln(4)} + \frac{1}{2_1} \cdot \frac{2}{1}$

$\Rightarrow \frac{\ln(30)}{2\ln(4)} + 2$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

$F(t) = ae^{-kt} + c$

Given: when $t=0$, $F(0) = 200^\circ F$

when $t=2$, $F(2) = 190^\circ F$

As $t \to \infty$, $F(t) = 75^\circ F$ and

As $t \to \infty$, $F(t) = C$

Step 1: Find c,

$\boxed{c = 75^\circ F}$

Step 2: Find a,

$F(0) = a + c$

$200 = a + 75$

$a = 200 - 75$

$\boxed{a = 125}$

Step 3: Find k,

$F(t) = ae^{-kt} + c$

$F(2) = 125e^{-k(2)} + 75$

$F(2) = 125e^{-2k} + 75$

$190 = 125e^{-2k} + 75$

$190 - 75 = 125e^{-2k}$

$115 = 125e^{-2k}$

$e^{-2k} = \frac{115}{125} = \frac{23}{25}$

$e^{-2k} = \frac{23}{25}$

$\ln(e)^{-2k} = \ln\left(\frac{23}{25}\right)$

$-2k = \ln\left(\frac{23}{25}\right)$

$k = \frac{\ln\left(\frac{23}{25}\right)}{-2}$

$k = \frac{\ln\left(\frac{25}{23}\right)}{2}$

$k \approx 0.0417$

$\boxed{F(t) = 125e^{-0.0417t} + 75}$

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: Nunez Valdivia, Alexa WA3289612

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$a = 27.03$ $\quad b = 1.67$ $\quad r = 0.67$

$1 + r = 1.67$

$r = 0.67$

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

$a = 12.75$

$r = 4.1\% = 0.041$

$b = 1 + 0.041$

$p(t) = 12.75(1.041)^t$

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

$p(25) = 12.75(1.041)^{25}$

$= 34.81$

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

$P(t) = 6500 - 60t$

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

$5\% = 0.05$ $\qquad P(t) = 6500(1 - 0.05)^t$

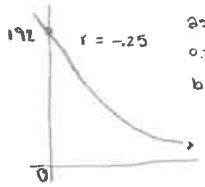
$P(t) = 6500(0.95)^t$

4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

$y = 0$

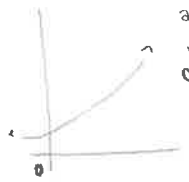
5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$ 0



as $x \to \infty$, 0.75 will become smaller but never under 0, it approaches 0

(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$ 0



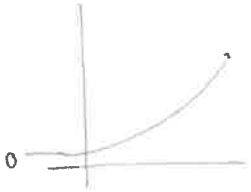
as $t \to -\infty$, it will approach 0, but never pass it.

(c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to$ 1



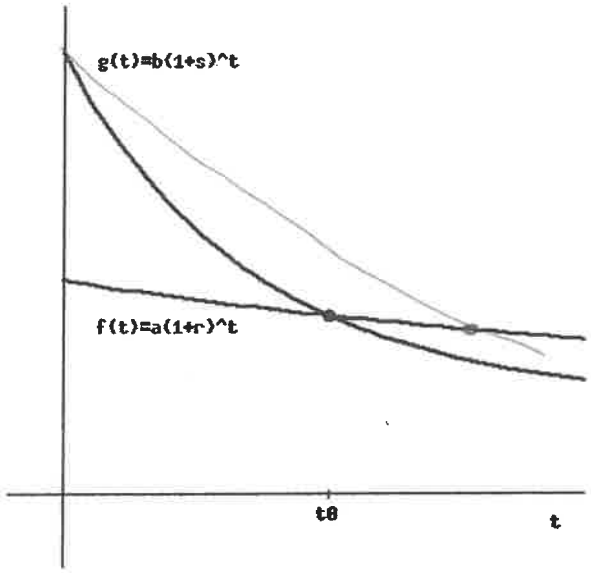
approaches 1 since it won't go less than 0.

(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$ 0



approaches 0 but never reaches it, since the exponent is less than 1.

6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.



If s is increased, t_0 would increase.

7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$m(0) = 160e^{-0.025(0)}$ $\quad$ $\boxed{m(0) = 160}$

$= 160e^{0}$ $\quad = 160 \cdot 1 = 160$

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

$m(10) = 160e^{-0.025(10)}$

$\boxed{= 160e^{-0.25}}$

$\boxed{\approx 124.6081}$

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

   (a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

   the order matters when there is multiple types of transformations, (scaling, reflecting, shifting). This problem only has a horizontal and vertical shift so it shouldn't matter in this case.

   1st : horizontal shift right by 4. (e^{x-4}) 2nd: shift up by 2. $(e^{x-4}+2)$

   (b) (4 pts) State the domain and range of $f(x)$.

   domain : $\mathbb{R}$, $(-\infty, \infty)$

   range : $(2, \infty)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

   (a) (4 pts) $\ln 10 = y$

   $e^y = 10$

   (b) (4 pts) $\ln x = 5$

   $e^5 = x$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

    (a) (4 pts) $\ln(e^{1.7})$

    $\ln(e^{1.7}) = 1.7$

    (b) (4 pts) $e^{\ln 4}$

    $e^{\ln 4} = 4$

    (c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

    $\ln\left(\frac{1}{e^3}\right) = \ln(e^{-3}) = -3$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

$$\ln(e^{-3x+2}) = \ln(10)$$

$$-3x + 2 = \ln(10)$$

$$-2 \qquad -2$$

$$\frac{-3x}{-3} = \frac{\ln(10) - 2}{-3}$$

$$\boxed{x = \frac{\ln(10) - 2}{-3}}$$

(b) (4 pts) $8 - \ln(2 - x) = 1$.

$$8 - \ln(2-x) = 1$$

$$-8 \qquad -8$$

$$-\ln(2-x) = -7$$

$$\ln(2-x) = 7$$

$$e^{\ln(2-x)} = e^7$$

$$2 - x = e^7$$

$$-x = e^7 - 2$$

$$\boxed{x = 2 - e^7}$$

(c) (4 pts) $4^x = 3$

$$\ln(4^x) = \ln(3)$$

$$x \cdot \ln(4) = \ln(3)$$

$$\boxed{x = \frac{\ln(3)}{\ln(4)}}$$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$$\frac{2 \cdot 4^{2x-4}}{2} = \frac{60}{2}$$

$$4^{2x-4} = 30$$

$$\ln(4^{2x-4}) = \ln(30)$$

$$\frac{2x-4 \cdot \ln(4)}{2 \cdot \ln(4)} = \frac{\ln(30)}{2 \cdot \ln(4)}$$

$$x - 4 = \frac{\ln(30)}{2 \cdot \ln(4)}$$

$$+4 \qquad +4$$

$$\boxed{x = \frac{\ln(30)}{2 \cdot \ln(4)} + 4}$$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

$F(0) = 200^\circ F$

$c = 75^\circ$

$a = 125$

$k = -\frac{1}{2} \cdot \ln\left(\frac{115}{125}\right)$

$F(2) = 190^\circ$

$a + c = 200$

$a + 75 = 200$

$a = 125$

$190 = 125e^{-k(2)} + 75$

$-75 \quad\quad -75$

$\frac{115}{125} = \frac{125e^{-k(2)}}{125}$

$\frac{115}{125} = e^{-k(2)}$

$\ln\left(\frac{115}{125}\right) = \ln\left(e^{-k(2)}\right)$

$\ln\left(\frac{115}{125}\right) = -k \cdot 2$

$-\frac{1}{2} \cdot \ln\left(\frac{115}{125}\right) = k$

$k \approx 0.04169$

formula:

$$F(t) = 125e^{\frac{1}{2} \cdot \ln\left(\frac{115}{125}\right)t} + 75$$

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: Lwin, Kyaw Wai WA32653997

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a =$ 27.03 $\quad b =$ 1.67 $\quad r =$ 0.67 67 %

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

rate 4.1% → 104.1% = 1.041

$p(t) = 12.75(1.041)^t$

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

$t = 25$, $p(25) = 12.75(1.041)^{25}$

$= 12.75(2.730)$

$= 34.80$

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

$p(t) = 6500 - 60t$

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

$p(t) = 6500(0.95)^t$

4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$ Approaches zero as x ~~increases~~ ∞

let $x = 1 \to 192(0.75)^1 = 144$

$x = 5 \to 192(0.75)^5 = 45$

$x = 10 \to 192(0.75)^{10} = 10.8$

(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$ Approaches zero as t ~~increases~~ ∞

let $t = -1, 2(3)^{-1} = \frac{2}{3} \approx 0.667$

$t = -5, 2(3)^{-5} = \frac{2}{243} \approx 0.0082$

$t = -10, 2(3)^{-10} = \frac{2}{59049} \approx 0.00003387$

(c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to$ Approach ~~is~~ to 15

let $t = 1, 5(3 + (0.9)^1) = 19.5$

$t = 5, 5(3 + (0.9)^5) = 17.9$

$t = 10, 5(3 + (0.9)^{10} = 16.7$

(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$ Approaches zero

let $t = -1, 8e^{0.5(-1)} \approx 4.852$

$t = -5, 8e^{0.5(-5)} \approx 0.657$

$t = -10, 8e^{0.5(-10)} \approx 0.0539$

6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.

let $s = (-0.2)$

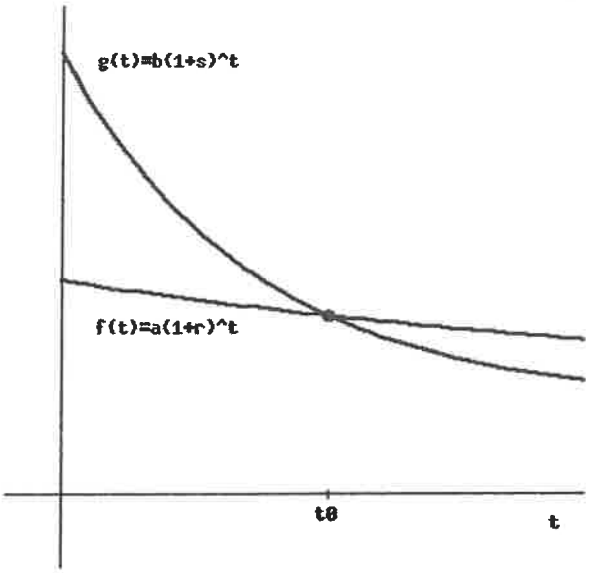
$g(t) = 10(1+0.2)^t$

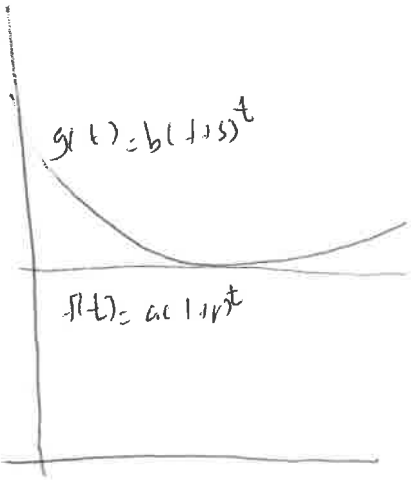
$= 10(0.8)^t$

$s = (-0.8)$

$g(t) = 10(1+(-0.8))^t$

$= 10(0.2)^t$





7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$m(t) = 160e^{-0.025t}$, $t = 0$, $m(0) =$ 160 gram

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

$t = 10$

$m(10) = 160e^{-0.025(10)}$

$\simeq 124.6081$ grams

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

   (a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

   subtract 4 from x
   shift up by 2

   does not matter

   (b) (4 pts) State the domain and range of $f(x)$.

   $e^{x-4} > 0$

   Domain $(-\infty, \infty)$
   Range $(2, \infty)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

   (a) (4 pts) $\ln 10 = y$

   $e^{(\ln 10)} = e^y$

   $10 = e^y$

   (b) (4 pts) $\ln x = 5$

   $e^{(\ln x)} = e^5$

   $x = e^5$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

    (a) (4 pts) $\ln(e^{1.7})$ $= 1.7$

    (b) (4 pts) $e^{\ln 4}$ $= 4$

    (c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$ $= \ln(e^{-3}) = -3$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

$$\ln e^{-3x+2} = \ln 10$$

$$-3x+2 = \ln 10$$

$$\boxed{x = \frac{2-\ln 10}{3}}$$

(b) (4 pts) $8 - \ln(2-x) = 1$.

$$\ln(2-x) = 7$$

$$e^{\ln(2-x)} = e^{7}$$

$$2-x = e^{7}$$

$$\boxed{x = 2-e^{7}}$$

(c) (4 pts) $4^x = 3$

$$x = \log_4 3$$

$$\boxed{x = \frac{\ln 3}{\ln 4}}$$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$$4^{2x-4} = 30$$

$$2x-4 = \log_4 30$$

$$2x-4 = \frac{\ln 30}{\ln 4}$$

$$\boxed{x = \frac{\ln 30}{2\ln 4} + 2}$$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

$F(t) = ae^{-kt} + c$

$200 = a + 75$

$a = 125$

$F(2) = 125\ e^{-k2} + 75$

$190 - 75 = 125e^{-k2}$

$115 = 125e^{-k2}$

$e^{-k2} = \frac{115}{125}$

$e^{-2k} = 0.92$

$\ln e^{-2k} = \ln 0.92$

$-k = \frac{\ln 0.92}{2}$

$k = \frac{0.0834}{2} \approx 0.0417$

$F(t) = 125e^{-(0.0417)t} + 75$

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: Garcia, Jocelyn, WA32167035

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a = 27.03$ $\quad b =$ _____ $\quad r = 1.67$

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

$Q = ab^t = a(1+4.1)^t$

$Q = 27.03 = 27.03(1+4.1)^t$

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

$P = 6500 - (60)^t$

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

$P = 6500 - (5)^t$

4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

there is no horizontal asymptote

5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$ ∞

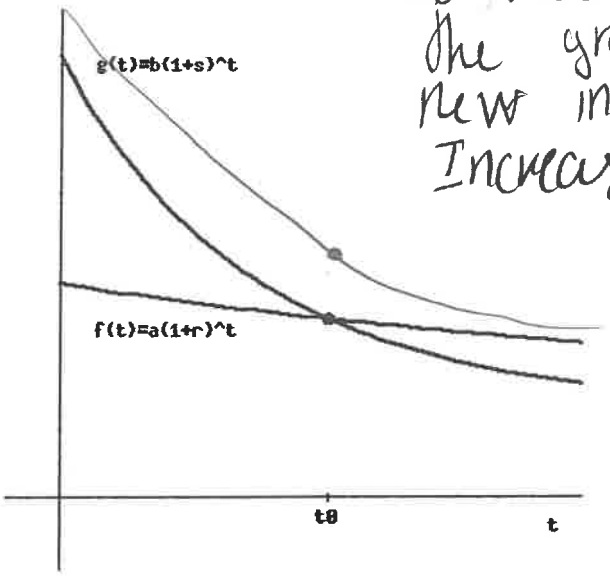
(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$ 0

(c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to$ ∞

(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$ ∞

6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.

To would move with the graphs whenever a new intersection is Increase



7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$m(0) = 160e^{-0.025} = 156.05$

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

$m(10) = 160e^{-0.025(10)}$

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

   (a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

   Shifting & scaling, yes order matters
   shift, then scale

   (b) (4 pts) State the domain and range of $f(x)$.

   D) $(-\infty, \infty)$
   R) $(0, \infty)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

   (a) (4 pts) $\ln 10 = y$

   $e^{\ln 10}$

   (b) (4 pts) $\ln x = 5$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

    (a) (4 pts) $\ln(e^{1.7})$

    1.7

    (b) (4 pts) $e^{\ln 4}$

    4

    (c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

    -3

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

(b) (4 pts) $8 - \ln(2-x) = 1$.

$$-\ln(2-x) = -7$$
$$\frac{-\ln(2-x)}{2-x} = \frac{-7}{2-x}$$
$$-\ln = \frac{7}{2-x}$$

(c) (4 pts) $4^x = 3$

$$\frac{4}{4}$$
$$x = \frac{3}{4}.$$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$$\frac{4^{2x-4}}{4} = \frac{60}{2}$$

$2x$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

$a = 125 \quad c = 75 \quad e$

$F(0) = 200 \Rightarrow 200 = ae^{-k(t)} + 75$

$a = 200 - 75 = 125$

$F(2) = 190 \Rightarrow 190 = 125 + 75 \Rightarrow 115 = 125e^{-2k}$

$115/125 \; e^{-2k} \quad \ln(115/125) = -2k$

$k = \frac{1}{2}\ln(-0.083)$

$F(t) = 125e - \ln(115/121\ 8/12t + 75)$

$\frac{125}{115}t + 75 \qquad \frac{115}{125} = e^{-2k}$

$\frac{2}{\ln\left(\frac{125}{115}\right)} = \ln e^{-2k}$

$\ln\left(\frac{115}{125}\right) = \frac{-2k}{2} \quad k = 0.083$

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: Lo, Cayla

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a = 27.03$ $b = 1.67$ $r = 0.67$

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

$p(t) = 12.75(1+0.041)^t$

$p(t) = 12.75(1.041)^t$

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

$p(25) = 12.75(1.041)^{25}$

≈ 34.82

\$34.82

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

$P(t) = 6500 - 60t$

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

$P(t) = 6500(1-0.05)^t$

$P(t) = 6500(0.95)^t$

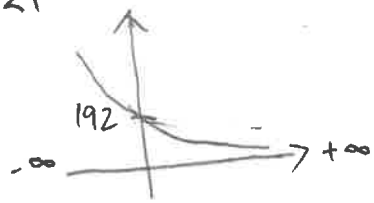
4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

The horizontal asymptote is $y=0$.

5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

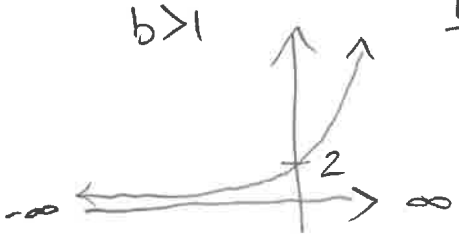
(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$ 0

$0<b<1$



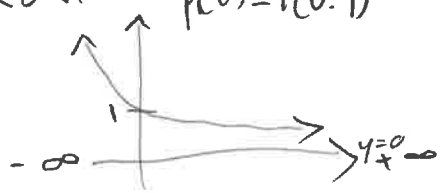
(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$ 0

$b>1$



(c) (4 pts) As $t \to \infty$, $5(3+(0.9)^t) \to$ 15

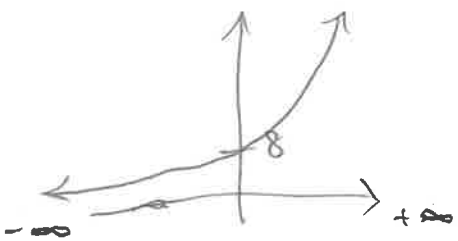
$0<b<1$ $p(t)=1(0.9)^t$



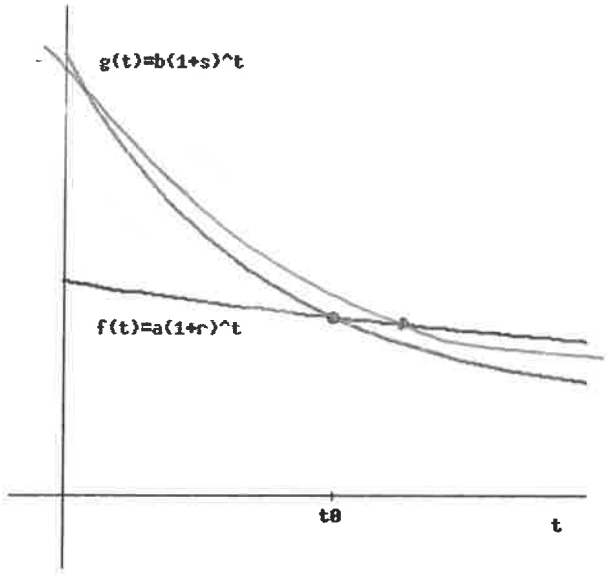
$f(t)=5(3+0)$
$=5\cdot 3$
$=15$

(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$ 0

$b>1$ $e \approx 2.7$



6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.



New g(t)

If the value of s is increased the value of to increased too.

7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$m(0) = 160e^{-0.025(0)}$

$= \boxed{160g}$

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

$m(10) = 160e^{-0.025(10)}$

$\approx \boxed{124.6081g}$

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

   (a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

   ① It horizontally shifted by 4 units.
   ② Then, vertically shifted by 2 units.

   (b) (4 pts) State the domain and range of $f(x)$.

   Dom: $(-\infty, \infty)$ Ran: $(-\infty, 2)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

   (a) (4 pts) $\ln 10 = y$

   $e^{\ln 10} = e^y$

   $\boxed{10 = e^y}$

   (b) (4 pts) $\ln x = 5$

   $e^{\ln x} = e^5$

   $\boxed{x = e^5}$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

    (a) (4 pts) $\ln(e^{1.7})$

    $\ln(e^{1.7}) = \boxed{1.7}$

    (b) (4 pts) $e^{\ln 4}$

    $e^{\ln 4} = \boxed{4}$

    (c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

    $\ln\left(\frac{1}{e^3}\right)$

    $= \ln(e^{-3})$

    $= \boxed{-3}$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

$-3x+2 = \ln(10)$

$-3x = \ln(10) - 2$

$x = \dfrac{\ln(10)-2}{-3}$

(b) (4 pts) $8 - \ln(2-x) = 1$.

$-\ln(2-x) = 1-8$

$-\ln(2-x) = -7$

$\ln(2-x) = 7$

$2-x = e^7$

$-x = e^7 - 2$

$x = -e^7 + 2$

$x = 2 - e^7$

(c) (4 pts) $4^x = 3$

$x \cdot \ln(4) = \ln(3)$

$x = \dfrac{\ln(3)}{\ln(4)}$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$4^{2x-4} = \dfrac{60}{2}$

$(2x-4) \cdot \ln(4) = \ln(30)$

$2x - 4 = \dfrac{\ln(30)}{\ln(4)}$

$2x = \dfrac{\ln(30)}{\ln(4)} + 4$

$x = \dfrac{\frac{\ln(30)}{\ln(4)} + 4}{2}$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

Given:
- When $t=0$, $F(0)=200\,°F$
- When $t=2$, $F(2)=190\,°F$
- As $t\to\infty$, $F(t)\to 75°$ and As $t\to\infty$, $F(t)\to$

$C=75$ $\quad a+C=F(0)$

$a=125$ $\quad a+75=200$

$k=\frac{\ln(\frac{23}{25})}{-2}$ $\quad a=200-75$

$=125$

Solve for k:

$F(2)=125e^{-k\cdot 2}+75$

$190=125e^{-k\cdot 2}+75$

$190-75=125e^{-2k}$

$\frac{115}{125}=e^{-2k}$

$\ln(\frac{23}{25})=-2k$

$\frac{\ln(\frac{23}{25})}{-2}=k$

The formula is $F(t)=125e^{\frac{\ln(\frac{23}{25})}{2}t}+75$.

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: SANCHEZ, CODY SN013 #WA3281696

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a = 27.03$ $\quad b = 1.67$ $\quad r = 67\%$

$b - 1 = r$

$r \times 100 = \%$

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

$a = 12.75$

$b = \frac{4.1}{100} = 0.041$

$r = b + 1 = 1.041$

Thus,

$p(t) = 12.75(1 + 0.038)^t$

$\approx$ or $12.75(1.038)^t$

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

IF $p(t) = 12.75(1.038)^t$

$p(25) = 12.75(1.038)^{30} = \$32.39$

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

$*$ -60 per year = LINEAR

Thus, $P(t) = m(t) + b$

so, $P(t) = -60(t) + 6500$

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

$*$ 5% = EXPONENTIAL

so, $P(t) = a(1+r)^t$

so, $P(t) = 6500(1 - 0.05)^t$

4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

$+\infty \to +$ is increasingly positive

Thus, $\lim_{t \to \infty} ab^t = 1$

5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$

$192(0.75)^x$

$b = (0.75)$, since $0 < 0.75 < 1$,

The $F(x)$ decays... so, $\lim_{x \to \infty} 192(0.75)^x = 0$

(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$

$b = 3$, $0 < 3 > 1$, The $F(x)$ grows

$\lim_{t \to -\infty} 2(3)^t = 0$

(c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to$

$5(3 + (0.9)^t = 5(3.9)^t \approx 19.5\ldots$

Thus, $\lim_{t \to \infty} 5(3.9)^t = 19.5$

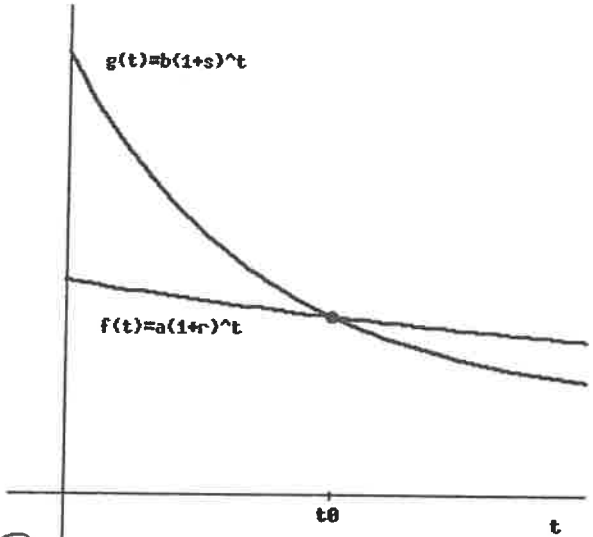
(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$

$b = e$, $e > 1$, $e^{0.5t} \to 0$

so $8(0) = 0$

Thus, $\lim_{t \to -\infty} 8e^{0.5t} = 0$

6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.



* NOTE that the functions are decreasing. Growth factor is $0 \leftrightarrow 1$. If Yo increase "s", g(t) decreases slowly. Thus t_0 increases

7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$t=0$, $m(0) = 160e^{-0.025(0)}$,

$e^0 = 1$, $160(1) =$ 160 grams

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

$m(10) = 160\, e^{-0.025(10)}$

$-0.025(10) = -0.25$; $160\, e^{-0.25}$

$160e^{-0.25} =$ 124.6081 grams

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

(a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

- we see "e^{x-4}", vs "e^x", & we see "+2"
- "+2" scales graph ↑ by 2 units
- NO, ORDER DON'T MATTER.

(b) (4 pts) State the domain and range of $f(x)$.

$D = (-\infty, \infty)$, $x \in \mathbb{R}$

$R = (2, \infty)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

(a) (4 pts) $\ln 10 = y$

$\ln b = a \longleftrightarrow e^a = b$

$\ln 10 = x$, so $\boxed{e^x = 10}$

(b) (4 pts) $\ln x = 5$

$\boxed{e^5 = x}$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

(a) (4 pts) $\ln(e^{1.7})$ ln & e = inverses...

$\ln(e^{1.7}) = \boxed{1.7}$

(b) (4 pts) $e^{\ln 4}$

$e^{\ln 4} = \boxed{4}$

(c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

$\frac{1}{e^3} = e^{-3}$

Thus, $\ln(e^{-3}) = \boxed{-3}$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

1.) TAKE ln of both sides

$\ln(e^{-3x+2}) = \ln(10)$

2.) $\ln(e^a) = a$

Thus, $-3x + 2 = \ln(10)$

3.) $-3x = \ln(10) - 2$ $\quad x = \dfrac{\ln(10) - 2}{-3}$

(b) (4 pts) $8 - \ln(2 - x) = 1$.

$-\ln(2-x) = -7$

$\rightarrow \ln(2-x) = 7 \quad (x = 1)$

$[\ln A = B \leftrightarrow e^B = A]$

$\ln(2-x) = 7, \; e^7 = 2 - x$: $\quad x = 2 - e^7$

(c) (4 pts) $4^x = 3$

1.) TAKE ln: $\ln(4^x) = \ln(3)$

2.) bring "x" down: $x \ln(4) = \ln(3)$

$$x = \frac{\ln(3)}{\ln(4)}$$

$\star \; 4^{2-4} = 16$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$2 \cdot 4^{2x-4} = 60$

$4^{2x-4} = 30 \quad (\div \text{ by } 2)$

$\ln(4^{2x-4}) = \ln(30)$

$\ln(4x-4) = \ln(30^4)$

$(4x-4) = \dfrac{4\ln(30)}{\ln(4)}$: $\quad 4x = 4 + \dfrac{4\ln(30)}{\ln(4)}$

$$x = 4 + \frac{\ln(30)}{\ln(4)}$$

$\frac{115}{135} = 0.85185$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

$c = 75°$

Thus, $F(t) = ae^{-k(t)} + 75°$

AT $F(0)$, Liquid $= 200°$

Thus $200 = ae^{-k(0)} + 75$

$e^{-k(0)} = 1$; $210 = a + 75$, $\begin{array}{r} 210 \\ -75 \\ \hline \end{array} = 135\,(a)$

Now, $F(t) = 135e^{-k(t)} + 75$

$F(2) = 190°$

so, $190 = 135\,e^{-k(2)} + 75$

$115 = 135\,e^{-k(2)} \cdot (-75)$

$\frac{115}{135} = e^{-k(2)} \quad (\div 135)$

$\ln\left(\frac{115}{135}\right) = \ln(e^{2k}) : \ln\left(\frac{115}{135}\right) = 2(k)$

$k = \frac{\ln\left(\frac{115}{135}\right)}{2}$ Thus, $F(t) = 135e^{\frac{\ln\left(\frac{115}{135}\right)}{2}t} + 7$

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: Ryan Angel WA2998374

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a =$ 27.03 $\quad b =$ 1.67 $\quad r =$ 67%

$1.67 - 1 = 0.67$

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

$p(t) = 12.75(1.041)^t$

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

$p(25) = 12.75(1.041)^{25} \approx 34.82$

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

$P(t) = 6500 - 60t$

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

$P(t) = 6500(0.95)^t$

$1 - 0.05 = 0.95$

4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

$y = 0$

5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$ 0

(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$ 0

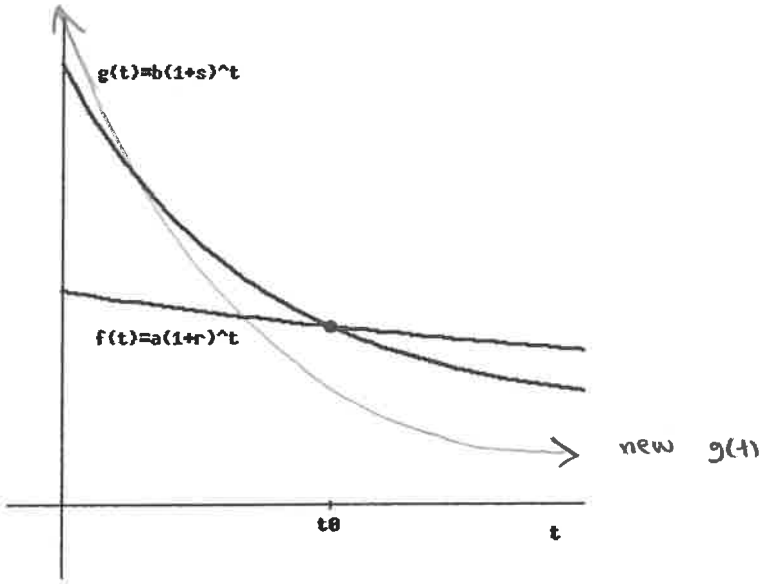
(c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to$ 15

$5(3 + (0)^t) = 5(3) = 15$

(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$ 0

$e \approx 2.7$

6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.



t_0 decreases because both graphs are decreasing. If only the value of s were to increase, the graph would decrease at a faster rate.

7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$m(0) = 160e^{-0.025(0)}$ $\quad m(0) = 160e^{0}$ $\quad m(0) = 160(1)$

$m(0) = 160$ grams

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

$m(15) = 160e^{-0.025(15)}$ $\quad \approx 109.9603$ grams

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

   (a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

usually, order matters, in this case, order does not matter because whether you shift up by 2 first, or right by 4 first, you'll still end up in the same spot because there is no reflecting or stretching / shrinking.

horizontal shift to the right by 4 units
vertical shift up by 2 units.

   (b) (4 pts) State the domain and range of $f(x)$.

domain: $(-\infty, \infty)$

range: $(2, \infty)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

   (a) (4 pts) $\ln 10 = y$

$e^y = 10$

   (b) (4 pts) $\ln x = 5$

$e^5 = x$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

   (a) (4 pts) $\ln(e^{1.7})$

1.7

   (b) (4 pts) $e^{\ln 4}$

4

   (c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

$\ln(e^{-3}) = -3$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

$-3x + 2 = \ln(10)$

$-2 \quad -2$

$\frac{-3x = \ln(10) - 2}{-3}$

$\boxed{\frac{\ln(10) - 2}{-3} = x}$

(b) (4 pts) $8 - \ln(2 - x) = 1$.

$+\ln(2-x) \quad +\ln(2-x)$

$8 = 1 + \ln(2-x)$

$-1 \quad -1$

$7 = \ln(2-x)$

$e^7 = 2 - x$

$+x \quad +x$

$x + e^7 = 2$

$-e^7 \quad -e^7$

$\boxed{x = -e^7 + 2}$

(c) (4 pts) $4^x = 3$

$\frac{x \cdot \ln(4) = \ln(3)}{\ln(4)}$

$\boxed{x = \frac{\ln(3)}{\ln(4)}}$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$\frac{}{2}$

$4^{2x-4} = 30$

$\frac{2x - 4 \cdot \ln(4) = \ln(30)}{\ln(4)}$

$2x - 4 = \frac{\ln(30)}{\ln(4)}$

$+4 \quad +4$

$2x = \frac{\ln(30)}{\ln(4)} + 4$

$\frac{}{2}$

$\boxed{x = \frac{\ln(30)}{2\ln(4)} + 2}$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

$200 = ae^{0} + 75$

$200 - 75 = 125$

$f(t) = 125e^{-kt} + 75$

$190 = 125e^{-k(2)} + 75$

$-75 \qquad -75$

$\frac{115}{125} = \frac{125e^{-k(2)}}{125}$

$\frac{115}{125} = e^{-k(2)}$

$\ln\left(\frac{115}{125}\right) = -k(2)$

$\frac{\ln(115/125)}{-2} = k$

$k \approx 0.041690$

$F(t) = 125e^{-0.041690t} + 75$

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: MYAZ, JEFFERY, WA3775177

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a = 27.03$ $\quad b = 1+r$ $\quad r = 0.0167$ %

$1+r = 1.67$

$r = 1.67 - 1$

$b = 0.67$

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

$r = 4.1$

$p(t) = 12.75(4.1)^t$

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

$p(t) = 12.75(4.1)^{25}$

$= 12.75(2.087)$

$= 2.66$ ~~dollars~~

2 dollars and 66 cents

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

$p(t) = 6500 - 60t$

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

$p(t) = 6500(0.95)^t$

4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

$y = 0$

5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

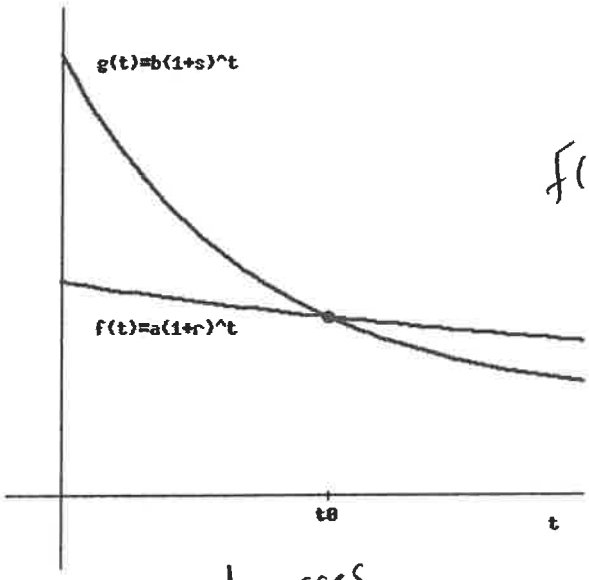
(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$ 0

(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$ 0

(c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to$ 15

(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$ 0

6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.



$f(t) = a(1+r)^t$

$g^{-t} = b(1+s)^{-t}$

for

t_0 decreases

7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$m(0) = 160e^{-0.025(0)}$

$= 160$

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

$m(10) = 160e^{-0.025(10)}$

$= 160e^{-0.25}$

$= 160 \frac{1}{e^{0.25}}$

$= \frac{160}{1.28} = 125$

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

   (a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

$g(-x) = e^{-x}$, the transformations doesn't matter

   (b) (4 pts) State the domain and range of $f(x)$.

domain $= (-\infty, \infty)$

range $(-2, \infty)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

   (a) (4 pts) $\ln 10 = y$

$10 = e^y$

   (b) (4 pts) $\ln x = 5$

$x = e^5$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

   (a) (4 pts) $\ln(e^{1.7})$ $= 1.7$

   (b) (4 pts) $e^{\ln 4}$ $= \ln 4$

   (c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$ $= -3$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

$$e^{-3x+2} = 10$$
$$-3x+2 = \ln 10$$
$$-3x = \ln 10 - 2$$
$$3x = 2 - \ln 10$$
$$x = \frac{2 - \ln 10}{3}$$

(b) (4 pts) $8 - \ln(2 - x) = 1$.

$$8 - \ln(2-x) = 1$$
$$-\ln(2-x) = -7$$
$$\ln(2-x) = 7$$
$$2 - x = \ln 7$$
$$-x = \ln 7 - 2$$
$$x = 2 - \ln 7$$

(c) (4 pts) $4^x = 3$

$$4^x = 3$$
$$\ln 4^x = \ln 3$$
$$x \ln 4 = \ln 3$$
$$x = \frac{\ln 3}{\ln 4}$$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$$2 \cdot 4^{2x-4} = 60$$
$$4^{2x-4} = 30$$
$$2x - 4 \ln 4 = \ln 30$$
$$2x - 4 = \frac{\ln 30}{\ln 4}$$
$$2x = \frac{\ln 30}{\ln 4} + 5$$
$$x = \frac{\ln 30}{2 \ln 4} + \frac{5}{2}$$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

$c = 75$

$F(t) = ae^{-kt} + 75$

$F(0) = 200$

$200 = ae^{-k \cdot 0} + 75$

$125 = a$

$a = 125$

$F(t) = 125e^{-kt} + 75$

$F(2) = 190$

$190 = 125e^{-2k} + 75$

$125e^{-2k} = 115$

$e^{-2k} = \frac{115}{125}$

$-2k = \ln\left(\frac{115}{125}\right)$

$2k = -\ln\left(\frac{23}{25}\right)$

$k = \frac{-\ln\left(\frac{23}{25}\right)}{2}$

$= \frac{\ln\left(\frac{25}{23}\right)}{2}$

$k = 0.0417$

$F(t) = 125e^{-0.0417t} + 75$

$F(t) = 0$

$125e^{-0.0417t} + 75 = 0$

$125e^{-0.0417t} = -75$

$e^{-0.0417t} = \frac{-75}{125}$

$\frac{1}{e^{0.0417t}} = \frac{75}{125}$

$-0.0417t = \ln\left(\frac{75}{125}\right)$

$0.0417t = \ln\left(\frac{125}{75}\right)$

$0.0417t = \ln\left(\frac{5}{3}\right)$

$t = \frac{\ln\left(\frac{5}{3}\right)}{0.0417}$

$t = \frac{0.511}{0.0417}$

$t = 12.25$ minutes

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

***

Last name, first name, ID#: Li Justin, S00021792

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a = 27.03$ $\quad b = 1.67$ $\quad r = 0.67$

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

$P(t) = 12.75(1+0.041)^t$

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

$P(25) = 12.75(1+0.041)^{25} = 34.82$

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

$P(t) = 6500 - 60t$

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

$P(t) = 6500(1+0.05)^t$

4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

As b gets smaller and smaller by t, it will eventually get close to 0

5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$

getting smaller



(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$

getting bigger

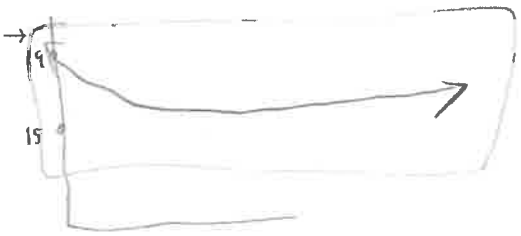


(c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to$

cap

start

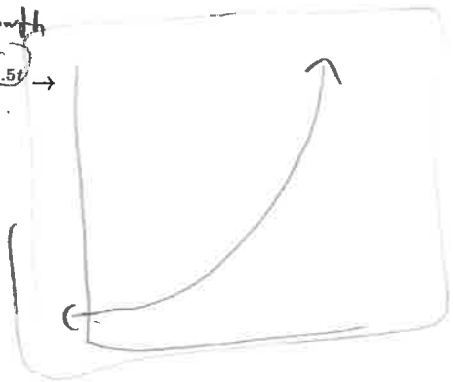
smaller



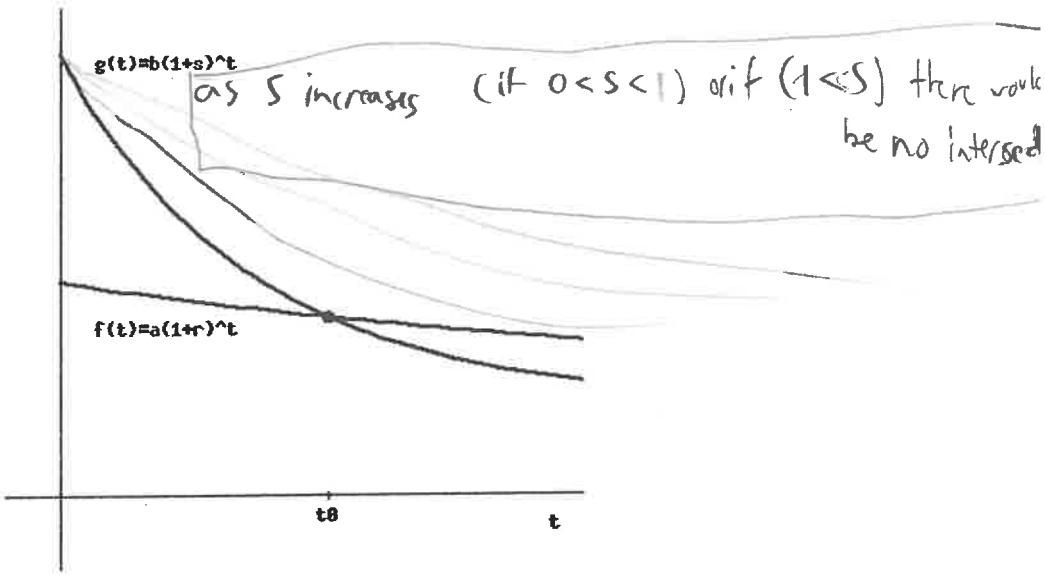
(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$

start

growth



6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.



as s increases (if $0<s<1$) or if $(1<s)$ then would be no intersect

7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$160e^{-0} = 160e$ or 434.93

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

$160e^{-0.25} = 124.6081$ or $45.8408e$

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

(a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

1. ↓ shift by 4 $= e^{x-4}$

2. another ↑ shift by 2 $= 2 + e^{x-4}$ or $e^{x-4} + 2$. The order may not m

(b) (4 pts) State the domain and range of $f(x)$.

↔ $d = (-\infty, \infty)$

↕ $r = (2, \infty)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

(a) (4 pts) $\ln 10 = y$?

(b) (4 pts) $\ln x = 5$

e e

↓

$x = e^5$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

(a) (4 pts) $\ln(e^{1.7})$

↳ $1.7 \cdot \ln(e) \rightarrow 1.7$

(b) (4 pts) $e^{\ln 4}$

↓

4

(c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

↳ $\ln(1) - \ln(e^3) = \ln(1) - 3$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

$\ln \downarrow \quad \downarrow \ln$

$-3x+2 = \ln(10)$

$-2 \quad\quad -2$

$-3x = \ln(10) - 2$

$\frac{-3x}{-3} = \frac{\ln(10)-2}{-3}$

$x = \frac{\ln(10)-2}{-3}$

(b) (4 pts) $8 - \ln(2-x) = 1$.

$-8 \quad\quad -8$

$\ln(2-x) = -7$

$\ln(2) \quad\quad +\ln(2)$

$\ln(-x) - \ln(2) = -7$

$\ln(-x) = -7 + \ln(2)$

$e \quad\quad e$

$-x = e^{-7+\ln(2)}$

$x = -e^{-7+\ln(2)}$

(c) (4 pts) $4^x = 3$

$\ln(4^x) = \ln(3)$

$x \cdot \ln(4) = \ln(3)$

$\frac{x \cdot \ln(4)}{\ln(4)} = \frac{\ln(3)}{\ln(4)}$

$x = \frac{\ln(3)}{\ln(4)}$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$\frac{2}{2} \quad\quad \frac{2}{2}$

$4^{2x-4} = 30$

$\ln \quad\quad \ln$

$(2x-4)\ln(4) = \ln(30)$

$\frac{(2x-4)\ln(4)}{\ln(4)} = \frac{\ln(30)}{\ln(4)}$

$2x-4 = \frac{\ln(30)}{\ln(4)}$

$+4 \quad\quad +4$

$2x = \frac{\ln(30)}{\ln(4)} + 4$

$\frac{2x}{2} = \frac{\frac{\ln(30)}{\ln(4)} + 4}{2}$

$x = \frac{\left(\frac{\ln(30)}{\ln(4)} + 4\right)}{2}$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

$F(0) = ae^{0} + 75$ $\quad a = \frac{200-75}{125}$

$125 \times 1 + 75$

$A = 125$

$C = 75$

$K \approx -0.0417$

$T = -1.(\ldots)-1$

$F(2) = 125e^{-k2} + 75 = 190$

$-75 \quad -75$

$\frac{125}{125} e^{-k2} = \frac{115}{125} \rightarrow 15 = \frac{23}{25}$

$e^{-k2} = \frac{23}{25}$

$\ln \quad \ln$

$\frac{-2k}{-2} K = \ln\left(\frac{23}{25}\right)$

$k = \frac{\ln\left(\frac{23}{25}\right)}{-2} = -0.0417$

$F(t) = 125e^{0.0417t} + 75$

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: Jadene Marquez WA3230981

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a =$ 27.03 $b =$ 1.67 $r =$ 25.36

Q=27.03 (starting value) 1.67 (growth factor) growth rate 25.36

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

a=12.75 starting price percentage 4.1% = 0.041 (30) years from today

\$25.50 12.75^2

t = 25.50

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

25.50 + 25.50 = 51.00 25.91

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

6500 t = 0 6500 - 60 = 6440 60% = 0.6

t = 3864

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

5% = 0.05 P(t) = 195

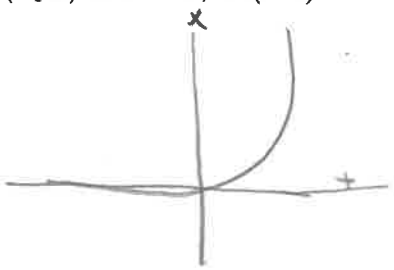
4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

$y = ab^t$ as $t \to \infty$

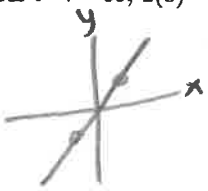
I'd say there is no horizontal line because it would be curved not horizontal

5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

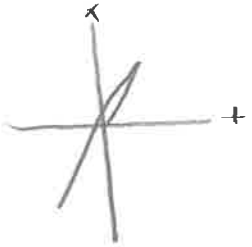
(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$ 144



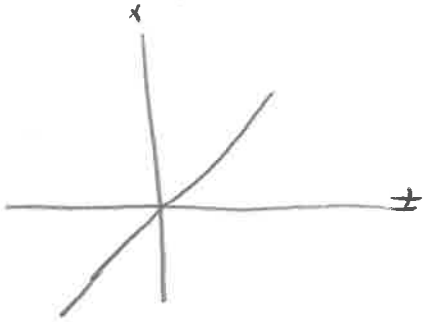
(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$ 6



(c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to$ 39

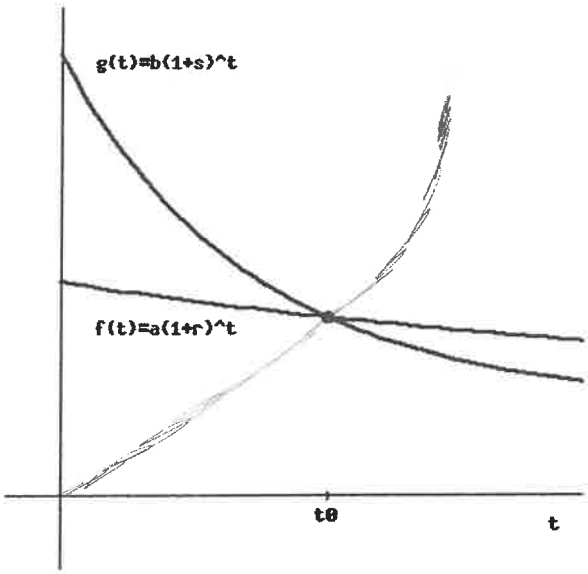


(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$ 2.82



6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.

$g(t)=b(1+s)^t$



7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

160 ^ -0.025

= 0.881

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

1408.51

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

   (a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

   $f(x) = e^{x-4} + 2$ $\quad g(x) = e^x$

   18 $\quad x = 20$

   (b) (4 pts) State the domain and range of $f(x)$.

   domain = R

   range = 20

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

   (a) (4 pts) $\ln 10 = y$

   2.30

   $b^{10} = 230$

   (b) (4 pts) $\ln x = 5$

   $b^5 = 160$

   1.60

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

    (a) (4 pts) $\ln(e^{1.7})$

    0.53 $\quad \ln(1.7)$

    (b) (4 pts) $e^{\ln 4}$

    1.38 $\quad \ln(4)$

    (c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

    $\ln\left(\frac{1}{3}\right)$

    -1.09

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

0.36

$e^{-3x+2} = 10$

0.036

(b) (4 pts) $8 - \ln(2 - x) = 1$.

$8 - \ln(2-x) = 1$

5.545

(c) (4 pts) $4^x = 3$

$4^x = 3$

64

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$2 \cdot 4^{2-4} = 6$

1106.75

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

200 /2 = 100 $\quad$ 100 + 90 = 190 $\quad$ 2.53 %

f(t) = ae^-kt + C $\quad$ cooling temp 2.53 %. $\quad$ 480.7

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: Jow, Anthony WA3277268

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a = 27.03$ $b = 1.67$ $r = 0.67$

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

$p(t) = 12.75(1.041)^t$

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

$p(25) = 12.75(1.041)^{25} \Rightarrow p(25) = 34.82$

\$34.82

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

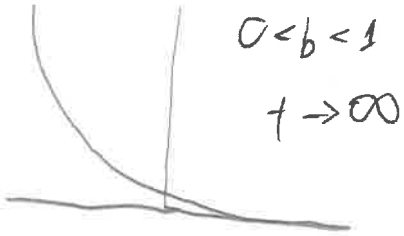
$P(t) = 6500 - 60t$

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

$P(t) = 6500(0.95)^t$

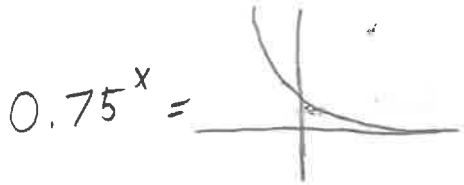
4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

$y \to 0$ as $t \to \infty$

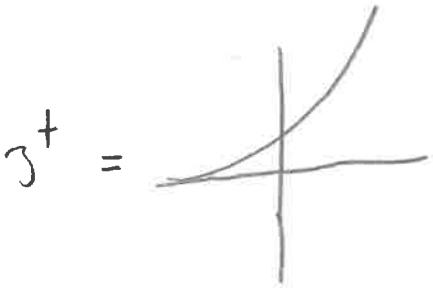


5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$ 0

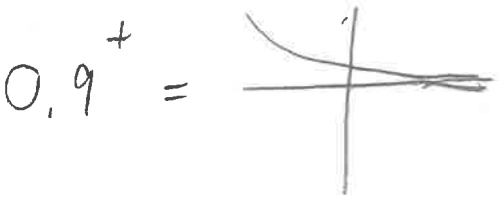


(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$ 0



(c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to$ 15

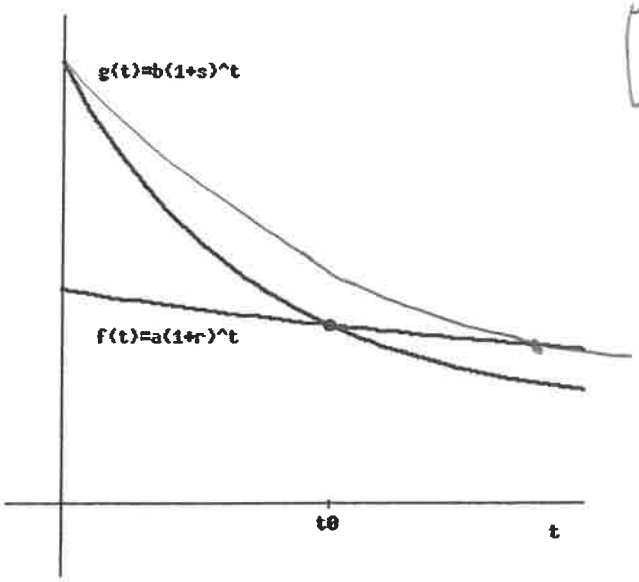
$5(3+0) = 15$



(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$ 0



6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.



Increase

7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$m(0) = 160e^{-0.025(0)} \Rightarrow 160e^{0} \Rightarrow 160(1) \Rightarrow 160$

160 g

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

$m(10) = 160e^{-0.025(10)} \Rightarrow 160e^{-0.250}$

$\Rightarrow 160(0.778800783)$

$\Rightarrow 124.6081$

124.6081 g

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

(a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

shift right by 4, then shift up by 2. Order does not matter in this case, but usually does.

(b) (4 pts) State the domain and range of $f(x)$.

Domain = All real numbers $(-\infty < x < \infty)$

Range = $(2 < y < \infty)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

(a) (4 pts) $\ln 10 = y \Rightarrow e^{\ln(10)} = e^y \Rightarrow 10 = e^y$

(b) (4 pts) $\ln x = 5 \Rightarrow e^{\ln(x)} = e^5 \Rightarrow x = e^5$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

(a) (4 pts) $\ln(e^{1.7})$

1.7

(b) (4 pts) $e^{\ln 4}$

4

(c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

-3

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$. $\Rightarrow -3x+2 = \ln(10) \Rightarrow -3x = \ln(10) - 2 \Rightarrow x = \frac{2-}{3}$

$$x = \frac{2 - \ln(10)}{3}$$

(b) (4 pts) $8 - \ln(2-x) = 1$. $\Rightarrow \ln(2-x) = 7 \Rightarrow 2-x = e^7 \Rightarrow -x = e^7 - 2$

$$x = 2 - e^7$$

(c) (4 pts) $4^x = 3 \Rightarrow \log_4(4^x) = \log_4(3) \Rightarrow x = \log_4(3)$

$$x = \log_4(3)$$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60 \Rightarrow 4^{2x-4} = 30 \Rightarrow 2x-4 = \log_4(30) \Rightarrow 2x = \log_4(30)$

$$x = \frac{\log_4(30) + 4}{2}$$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

$F(t) = ae^{-kt} + c$

$c = 75$

$a = 125$

$\Rightarrow F(t) = 125e^{-kt} + 75$

$\Rightarrow 190 = 125e^{-k(2)} + 75$

$\Rightarrow 115 = 125e^{-2k}$

$\Rightarrow \frac{115}{125} = e^{-2k}$

$\Rightarrow \ln\left(\frac{115}{125}\right) = -2k$

$\Rightarrow k = -\frac{1}{2} \cdot \ln\left(\frac{115}{125}\right)$

$$F(t) = 125e^{\frac{1}{2}\ln\left(\frac{115}{125}\right)t} + 75$$

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: Liu, Wilson

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a = 27.03$ $b = 1.67$ $r = .67$

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

$p(t) = 12.75(1.041)^t$

starting value, inflation

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

$p(25) = 12.75(1.041)^{25} \approx 34.81$ dollars in 25 years

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

$p(t) = 6500 - 60t$

initial, loss

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

$p(t) = 6500(.95)^t$

decrease per year

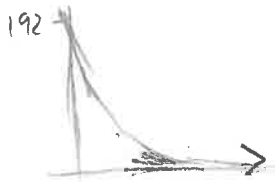
4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

$y=0$ approaches 0 the horizontal asymptote

5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

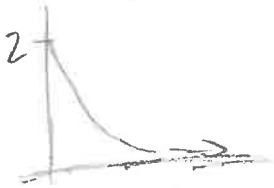
(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$

$0.75<1 \quad 192(0.75)^x \to 0$



(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$

$t \to -\infty, 2(3)^t \to 0$



(c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to$

$t \to \infty, 5(3+(0.9)^t) \to \infty$

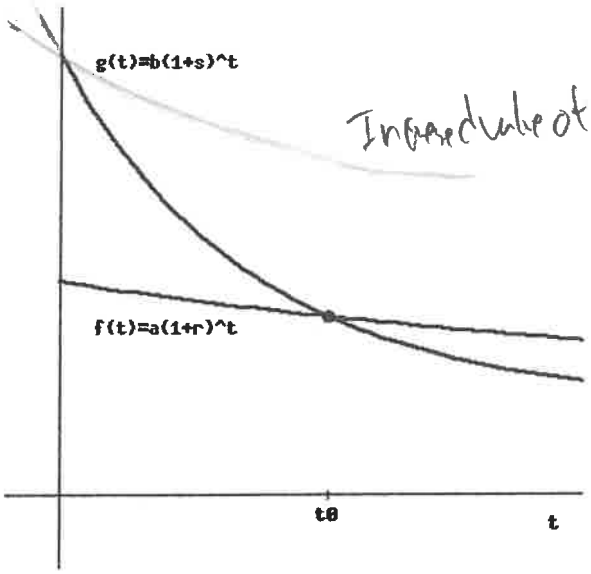


(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$

$t \to -\infty, 8e^{.5t} \to \infty$



6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.



Increase value of s

7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$160e^{-.025(0)} = 160e^{0} = 160$

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

$m(10) = 160e^{-.025(10)} = 160e^{-.25}$

$e^{-.25} \approx .7788$ $\quad 160 \cdot .7788 = 124.608$

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

   (a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

   - vertical shift up 2 units
   - horizontal shift left 4 units

   order does not matter as both are shifts.

   (b) (4 pts) State the domain and range of $f(x)$.

   D: $(-\infty, \infty)$

   R: $(2, \infty)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

   (a) (4 pts) $\ln 10 = y$

   $\ln 10 = y \rightarrow e^y = 10$

   (b) (4 pts) $\ln x = 5$

   $\ln x = 5 \rightarrow e^5 = x$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

    (a) (4 pts) $\ln(e^{1.7})$

    $\ln(e^{1.7}) = 1.7$

    (b) (4 pts) $e^{\ln 4}$

    $e^{\ln 4} = 4$

    (c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

    $\ln\left(\frac{1}{e^3}\right) = \ln(e^{-3}) = -3$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

$e^{-3x+2} = 10 \rightarrow -3x+2 = \ln(10) \rightarrow -3x = \ln(10) - 2 -$

$x = \frac{\ln(10)-2}{-3} \approx -.10086$

(b) (4 pts) $8 - \ln(2-x) = 1$.

$8 - \ln(2-x) = 1 \rightarrow \ln(2-x) = 8 \rightarrow 2-x = e^8 \rightarrow$

$x = 2 - e^8 \approx -2978.9579$

(c) (4 pts) $4^x = 3$

$4^x = 3 \rightarrow x\ln(4) = \ln(3) \rightarrow x = \frac{\ln(3)}{\ln(4)} \approx .792$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$2 \cdot 4^{2x-4} = 60 \rightarrow 4^{2x-4} = \frac{60}{2} \rightarrow (2x-4)\ln(4) = \ln(\frac{60}{2})$

$2x-4 = \frac{\ln(60/2)}{\ln(4)} \rightarrow 2x = \frac{\ln(60/2)}{\ln(4)} + 4 \rightarrow$

$x = \frac{\ln(60/2)}{2\ln(4)} + 2 \approx 3.2267$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

Initial temp = $200^\circ F$

Room temp = $75^\circ F$

A: $ae^0 + 75 = 200 \rightarrow a + 75 = 200 \rightarrow a = 125$

$F(t) = 125e^{-kt} + 75$

k: $125e^{-2k} + 75 = 190 \rightarrow 125e^{-2k} = 115 \rightarrow$

$e^{-2k} = 115/125 \rightarrow -2k = \ln\left(\frac{115}{125}\right) \rightarrow$

$k = -\frac{\ln(115/125)}{2} \approx .0416$

$F(t) = 125e^{-.0416t} + 75$

Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: Brambila Kristina

Please read the following instructions carefully.

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- You are welcome to ask me clarifying questions.

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$a =$ 27.03 $\quad b =$ 1.67 $\quad r =$ _____

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

$6500 - 60 = t + 60$

$+60$

$6500 = t + 60$

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

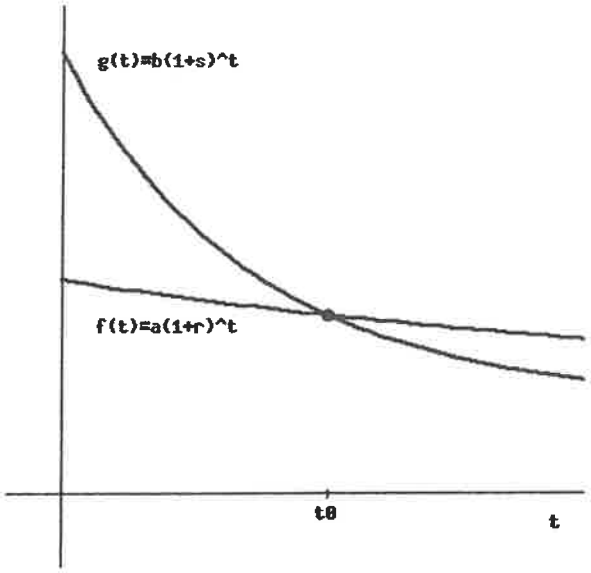
   (a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$

   (b) (4 pts) As $t \to -\infty$, $2(3)^t \to$

   (c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to$

   (d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$

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where $m(t)$ is measured in grams.

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$m(0) = 160e^{-0.025(0)}$

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

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   (b) (4 pts) State the domain and range of $f(x)$.

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   (a) (4 pts) $\ln 10 = y$

   (b) (4 pts) $\ln x = 5$

   log x 5

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

    (a) (4 pts) $\ln(e^{1.7})$

    log e1.7

    (b) (4 pts) $e^{\ln 4}$

    (c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

(b) (4 pts) $8 - \ln(2 - x) = 1$.

(c) (4 pts) $4^x = 3$

$4^x = 3$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$8/60$

$\frac{8^{2x-4}}{8} = \frac{60}{8}$

$2x - 4 = 7.5$

$+4 \quad +4$

$\frac{2x}{2} = \frac{11.5}{2}$

$x = \frac{11.5}{2}$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.